

MAADSBML AutoML Report For ARIMLABS

Generated On: 2026-02-13 16:42:36 (UTC)

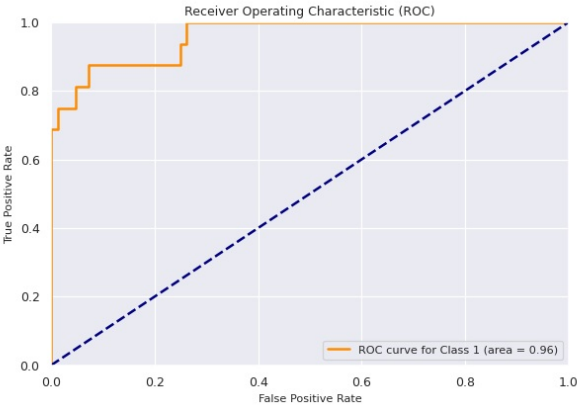
Best Model(s) Report For admin_maads_train_aug_binary_csv

MODEL DESCRIPTION

Model Trained On: 2026/02/13
Training Start Time: 1531
Training End Time: 1642
Was Data Normalized: Yes
Was Data Shuffled: Yes
Deep Analysis: Yes
Total Training Data Set: 1553
Training Data Percentage: 80%
Total Test Data Set: 387
Total # of Variables: 45
Adjusted for Seasonality: N
Total Algorithms Run: 1800
Removed Outliers: N
ROC AUC: 0.941
Precision: 0.700 (0.975 For Class=0)
Recall: 0.875 (0.929 For Class=0)
F1-Score: 0.778 (0.951 For Class=0)
Best Distribution FOR ACTUAL Y: [VONMISES](#)
Dependent Variable: LABEL
Independent Variables: ['customerage', 'transactionduration', 'loginattempts', 'txn_hour', 'is_night', 'is_weekend', 'mcc', 'high_risk_mcc', 'medium_risk_mcc', 'new_device', 'new_ip', 'customer_tenure_days', 'short_tenure', 'unusual_mcc_for_customer', 'txn_velocity_5min_sim', 'high_frequency', 'freq_2plus', 'proxy', 'transactiontype_Credit', 'transactiontype_Debit', 'transactiontype_nan', 'channel_Branch', 'channel_Online', 'channel_nan', 'customeroccupation_Doctor', 'customeroccupation_Engineer', 'customeroccupation_Retired', 'customeroccupation_Student', 'customeroccupation_nan', 'mcc_risk_low', 'mcc_risk_medium', 'mcc_risk_nan', 'country_BR', 'country_CA', 'country_DE', 'country_FR', 'country_GB', 'country_IN', 'country_MX', 'country_PL', 'country_SG', 'country_UA', 'country_US', 'country_nan']

Receiver Operating Characteristic Curve (ROC)

Using LogisticRegression algorithm for allseason



IMPORTANT FILE PATHS FOR RAW AND OUTPUT DATA

NOTE: These are DOCKER CONTAINER Paths. You can view these files inside the container by using the command: `docker exec -it {container id} bash` If you have re-run the container, these files will be GONE but they exist on your HOST machine. The HOST MACHINE location is based on the volumes you mapped when you ran the Docker container. The Docker RUN Volume Mappings are :: (For example here is the docker run command (use multiple -v for multiple mappings):

DOCKER RUN COMMAND: REFER TO [MAADSBML Documentation](#)

Docker Volume Mappings:

1. {HOST MACHINE FOLDER}/csvuploads:/maads/agentfilesdocker/dist/maadsweb/csvuploads:z
2. {HOST MACHINE FOLDER}/pdfreports:/maads/agentfilesdocker/dist/maadsweb/pdfreports:z
3. {HOST MACHINE FOLDER}/autofeatures:/maads/agentfilesdocker/dist/maadsweb/autofeatures:z
4. {HOST MACHINE FOLDER}/outliers:/maads/agentfilesdocker/dist/maadsweb/outliers:z
5. {HOST MACHINE FOLDER}/sqlloads:/maads/agentfilesdocker/dist/maadsweb/sqlloads:z
6. {HOST MACHINE FOLDER}/networktemp:/maads/agentfilesdocker/dist/maadsweb/networktemp:z
7. {HOST MACHINE FOLDER}/networks:/maads/agentfilesdocker/networks:z
8. {HOST MACHINE FOLDER}/exception:/maads/agentfilesdocker/dist/maadsweb/exception:z
9. {HOST MACHINE FOLDER}/staging:/maads/agentfilesdocker/dist/staging:z
10. {HOST MACHINE FOLDER}/backup:/Viperviz/viperviz/views/backup:z

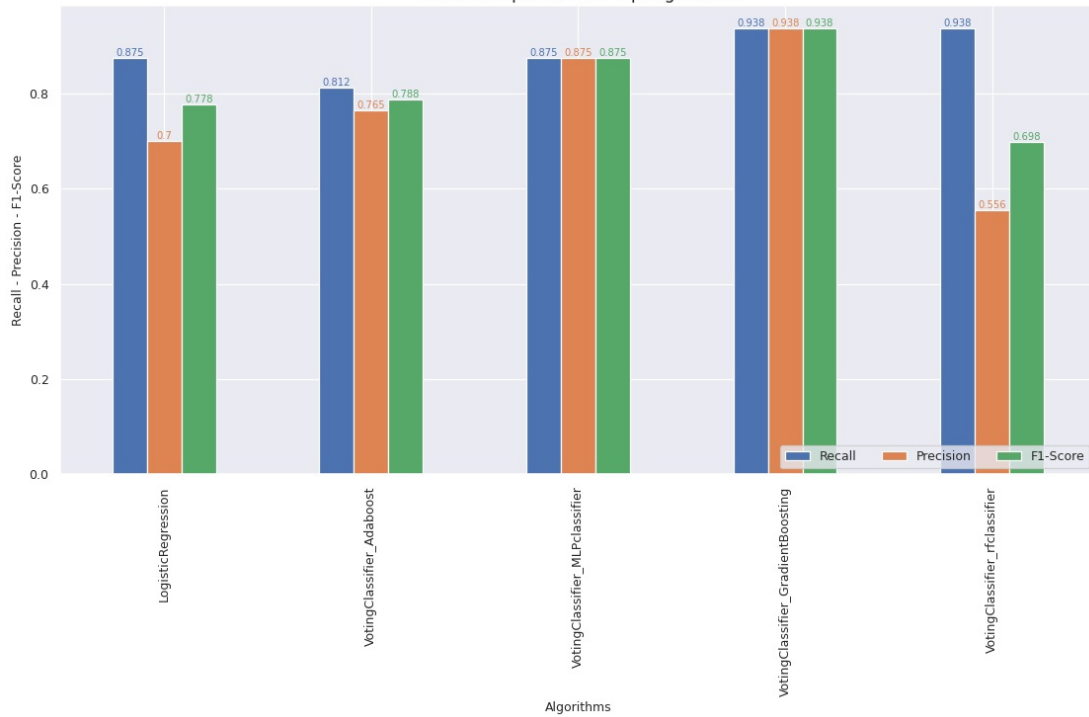
Path for Training Dataset File: [/maads/agentfilesdocker/dist/maadsweb/csvuploads/maads_train_aug_binary.csv](#)
Path for PDF Report (i.e. this file): [/maads/agentfilesdocker/dist/maadsweb/pdfreports/admin_maads_train_aug_binary_csv_no_seasons.pdf](#)
Path for AutoFeature File: [/maads/agentfilesdocker/dist/maadsweb/autofeatures/admin_maads_train_aug_binary_csv.csv](#)
Path for Outliers File: [/maads/agentfilesdocker/dist/maadsweb/outliers/admin_maads_train_aug_binary_csv.csv](#)
Path for Algo JSON File: [/maads/agentfilesdocker/dist/maadsweb/exception/admin_maads_train_aug_binary_csv_trained_algo_no_seasons.json](#)
Folder Path for MySQL Scripts: [/maads/agentfilesdocker/dist/maadsweb/sqlloads/](#)
Folder Path for Backup Reports: [/Viperviz/viperviz/views/backup/](#)
Path for Detailed Prediction File: [/maads/agentfilesdocker/dist/maadsweb/csvuploads/admin_maads_train_aug_binary_csv_prediction_details.csv](#)
Path for Algorithm Zip File (i.e pickle files): [/maads/agentfilesdocker/dist/maadsweb/networktemp/admin_maads_train_aug_binary_csv.zip](#)
Path for Algorithm Pickle Files:
1. [/maads/agentfilesdocker/networks/Otics_ADMIN_MAADS_TRAIN_AUG_BINARY_CSVALLEASON_AG1_4_LogisticRegression_normal_1553_ensembleone.pkl](#)
2. [/maads/agentfilesdocker/networks/Otics_ADMIN_MAADS_TRAIN_AUG_BINARY_CSVALLEASON_AG1_4_LogisticRegression_normal_1553_ensembleone_scalerx.pkl](#)

DESCRIPTIVE STATISTICS										
Variables	T-Statistic	Coefficients (alpha,beta)	Count	Mean	STD	MIN	25%	50%	75%	MAX
customerage	0.668	0.274, 0.0	1553.0	44.859	18.0	17.633	28.0	45.0	59.0	80.0
transactionduration	-0.954	0.308, -0.0	1553.0	119.505	10.0	70.221	63.0	113.0	162.0	300.0
loginattempts	-0.392	0.3, -0.007	1553.0	1.114	1.0	0.587	1.0	1.0	1.0	5.0
txn_hour	1.164	0.014, 0.017	1553.0	16.622	16.0	0.736	16.0	16.0	17.0	18.0
is_night	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
is_weekend	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
mcc	16.287	-0.564, 0.0	1553.0	5732.301	4111.0	1003.871	5311.0	5734.0	5944.0	7995.0
high_risk_mcc	29.621	0.089, 0.52	1553.0	0.328	0.0	0.47	0.0	0.0	1.0	1.0
medium_risk_mcc	-10.566	0.354, -0.244	1553.0	0.282	0.0	0.45	0.0	0.0	1.0	1.0
new_device	1.039	0.125, 0.167	1553.0	0.995	0.0	0.067	1.0	1.0	1.0	1.0
new_ip	1.82	-0.0, 0.293	1553.0	0.995	0.0	0.067	1.0	1.0	1.0	1.0
customer_tenure_days	-0.859	0.305, -0.0	1553.0	381.135	25.0	258.435	201.0	330.0	477.0	1858.0
short_tenure	0.32	0.291, 0.021	1553.0	0.023	0.0	0.148	0.0	0.0	0.0	1.0
unusual_mcc_for_customer	15.806	0.185, 0.328	1553.0	0.332	0.0	0.471	0.0	0.0	1.0	1.0
txn_velocity_5min_sim	9.053	0.15, 0.097	1553.0	1.41	1.0	0.92	1.0	1.0	1.0	5.0
high_frequency	5.829	0.268, 0.175	1553.0	0.126	0.0	0.331	0.0	0.0	0.0	1.0
freq_2plus	11.974	0.223, 0.276	1553.0	0.207	0.0	0.405	0.0	0.0	0.0	1.0
proxy	16.253	0.243, 0.554	1553.0	0.059	0.0	0.235	0.0	0.0	0.0	1.0
transactiontype_Credit	1.031	0.286, 0.025	1553.0	0.226	0.0	0.418	0.0	0.0	0.0	1.0
transactiontype_Debit	-1.031	0.311, -0.025	1553.0	0.774	0.0	0.418	1.0	1.0	1.0	1.0
transactiontype_nan	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
channel_Branch	-2.085	0.307, -0.046	1553.0	0.329	0.0	0.47	0.0	0.0	1.0	1.0
channel_Online	6.339	0.242, 0.134	1553.0	0.336	0.0	0.473	0.0	0.0	1.0	1.0
channel_nan	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
customeroccupation_Doctor	0.356	0.29, 0.009	1553.0	0.244	0.0	0.43	0.0	0.0	0.0	1.0
customeroccupation_Engineer	-2.264	0.305, -0.054	1553.0	0.269	0.0	0.444	0.0	0.0	1.0	1.0
customeroccupation_Retired	2.688	0.276, 0.064	1553.0	0.236	0.0	0.425	0.0	0.0	0.0	1.0
customeroccupation_Student	-0.781	0.296, -0.019	1553.0	0.25	0.0	0.433	0.0	0.0	1.0	1.0
customeroccupation_nan	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
mcc_risk_low	-16.804	0.412, -0.339	1553.0	0.39	0.0	0.488	0.0	0.0	1.0	1.0
mcc_risk_medium	-10.566	0.354, -0.244	1553.0	0.282	0.0	0.45	0.0	0.0	1.0	1.0
mcc_risk_nan	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
country_BR	5.146	0.284, 0.347	1553.0	0.015	0.0	0.123	0.0	0.0	0.0	1.0
country_CA	0.595	0.291, 0.057	1553.0	0.014	0.0	0.116	0.0	0.0	0.0	1.0
country_DE	3.109	0.288, 0.264	1553.0	0.011	0.0	0.104	0.0	0.0	0.0	1.0
country_FR	0.127	0.292, 0.016	1553.0	0.008	0.0	0.091	0.0	0.0	0.0	1.0
country_GB	-0.197	0.292, -0.019	1553.0	0.013	0.0	0.113	0.0	0.0	0.0	1.0
country_IN	-0.785	0.292, -0.092	1553.0	0.008	0.0	0.088	0.0	0.0	0.0	1.0
country_MX	1.387	0.29, 0.138	1553.0	0.012	0.0	0.11	0.0	0.0	0.0	1.0
country_PL	2.061	0.29, 0.21	1553.0	0.011	0.0	0.104	0.0	0.0	0.0	1.0
country_SG	-0.413	0.292, -0.042	1553.0	0.012	0.0	0.107	0.0	0.0	0.0	1.0
country_UA	2.076	0.289, 0.166	1553.0	0.013	0.0	0.113	0.0	0.0	0.0	1.0
country_US	-5.233	0.434, -0.162	1553.0	0.883	0.0	0.321	1.0	1.0	1.0	1.0
country_nan	nan	0.292, 0.0	1553.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
LABEL	NA	NA	1553.0	0.22	0.0	0.414	0.0	0.0	0.0	1.0

BEST ALGORITHM FOUND FOR THIS DATASET								
(Note: This trained model will be used to predict LABEL)								
Algorithm	Description	Model Results	ROC/AUC	Precision	Recall	F1-Score	Forecast Months	Season
LogisticRegression	Logistic Regression:	LogisticRegression(C=91.0852253511976, class_weight='balanced', fit_intercept=False, intercept_scaling=4.63866006216423, max_iter=314, random_state=43) Coefficient: [[-0.8707730414430451, -0.4759579322893768, 0.9523452048689657, -0.0289863004753869, 0.0, 0.0, 0.276694292568211, 0.2687990270097885, -1.5823216132011535, 1.9479550312270602, 2.548736476628011, -0.29043879231927233, -0.4432176327359184, 3.7441768535902322, 2.449782735972775, -0.4129184767914277, 1.2517251836484256, 3.809414970252244, -2.587148859839315, -2.4885236416855654, 0.0, 0.10180951658314877, -0.16980959490550607, 0.0, -1.2437966327954397, -1.4579680416225618, -1.0202221641776317, -1.3536856629292322, 0.0, -3.7621499153335103, -1.5823216132011535, 0.0, -0.4928751819165291, -0.3918187217077016, -0.29370860793189524, -0.06570207907850319, -1.645994158721242, -2.0235750186382773, 0.8630394785239865, 1.698297659242963, 0.431564177725161, -1.8847714064042749, -1.2701286426185887, 0.0]]	0.941	0.700: Class=1 (0.975: Class=0)	0.875: Class=1 (0.929: Class=0)	0.778: Class=1 (0.951: Class=0)	1 - 12	allseason

TOP 5 ALGORITHMS FOR ALLSEASON

Visual Comparisons of Top Algorithms

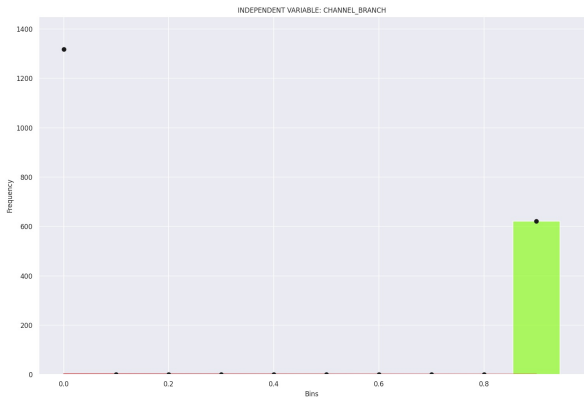


Num	Algorithm	Model ROC/AUC	Details	Season	Description
1	LogisticRegression	0.9410	Recall: 0.875 (class 1) Precision: 0.7 (class 1) F1 Score: 0.778 (class 1) Recall: 0.929 (class 0) Precision: 0.975 (class 0) F1 Score: 0.951 (class 0) False Negative Rate: 2.0% False Positive Rate: 6.0% True Negative Rate: 6.0% True Positive Rate: 14.0%	allseason	LOGISTIC REGRESSION:
2	VotingClassifier_Adaboost	0.9190	Recall: 0.812 (class 1) Precision: 0.765 (class 1) F1 Score: 0.788 (class 1) Recall: 0.952 (class 0) Precision: 0.964 (class 0) F1 Score: 0.958 (class 0) False Negative Rate: 3.0% False Positive Rate: 4.0% True Negative Rate: 4.0% True Positive Rate: 13.0%	allseason	ADA BOOST: ADA boost classifier
3	VotingClassifier_MLPclassifier	0.9180	Recall: 0.875 (class 1) Precision: 0.875 (class 1) F1 Score: 0.875 (class 1) Recall: 0.976 (class 0) Precision: 0.976 (class 0) F1 Score: 0.976 (class 0) False Positive Rate: 2.0% False Negative Rate: 2.0% True Negative Rate: 2.0% True Positive Rate: 14.0%	allseason	MULTIPLE LAYER PERCEPTRON CLASSIFIER: MLP Classifier
4	VotingClassifier_GradientBoosting	0.9180	Recall: 0.938 (class 1) Precision: 0.938 (class 1) F1 Score: 0.938 (class 1) Recall: 0.988 (class 0) Precision: 0.988 (class 0) F1 Score: 0.988 (class 0) False Positive Rate: 1.0% False Negative Rate: 1.0% True Negative Rate: 1.0% True Positive Rate: 15.0%	allseason	GRADIENT BOOSTING CLASSIFIER: Gradient Boosting Classifier
5	VotingClassifier_rfclassifier	0.8980	Recall: 0.938 (class 1) Precision: 0.556 (class 1) F1 Score: 0.698 (class 1) Recall: 0.857 (class 0) Precision: 0.986 (class 0) F1 Score: 0.917 (class 0)	allseason	RANDOM FOREST CLASSIFIER: Random forest

			False Negative Rate: 1.0% False Positive Rate: 12.0% True Negative Rate: 12.0% True Positive Rate: 15.0%		
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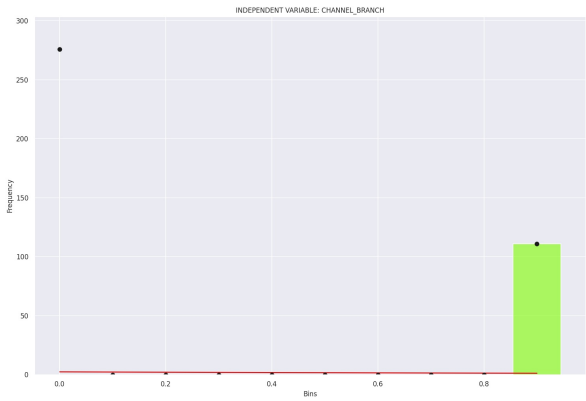
Detailed Histograms of Training and Test Data Sets

TRAINING VARIABLES



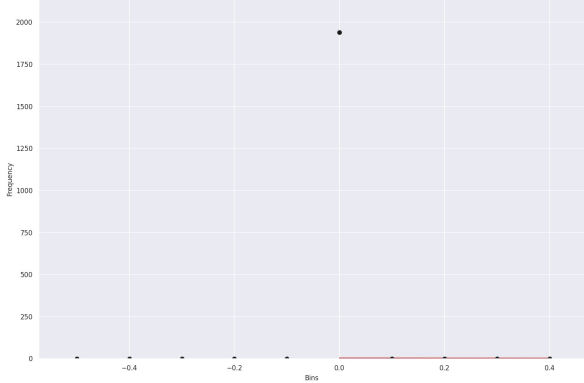
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1318	0	0	0	0	0
Share	68.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

TEST VARIABLES



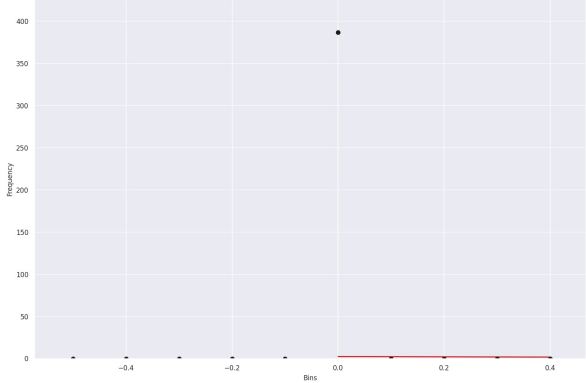
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	276	0	0	0	0	0
Share	71.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

INDEPENDENT VARIABLE: CHANNEL_NAN



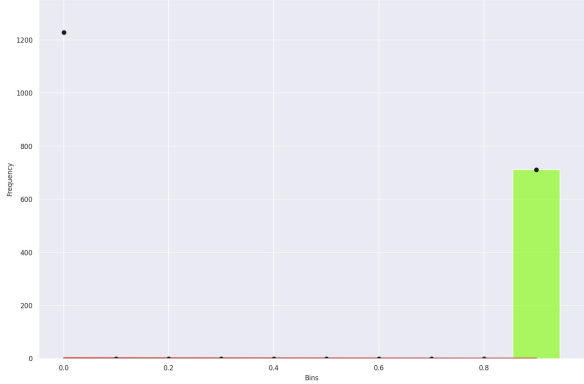
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

INDEPENDENT VARIABLE: CHANNEL_NAN

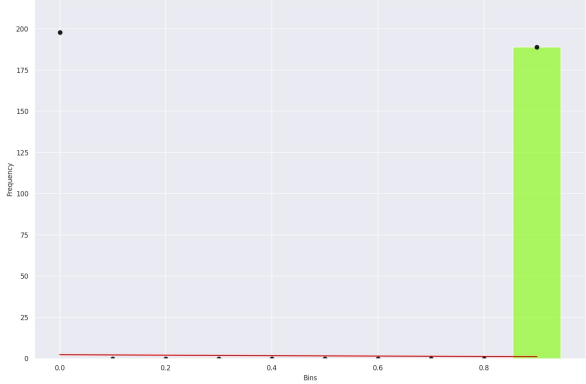


Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

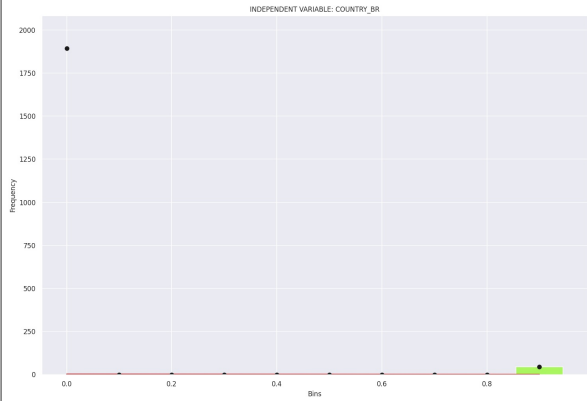
INDEPENDENT VARIABLE: CHANNEL_ONLINE



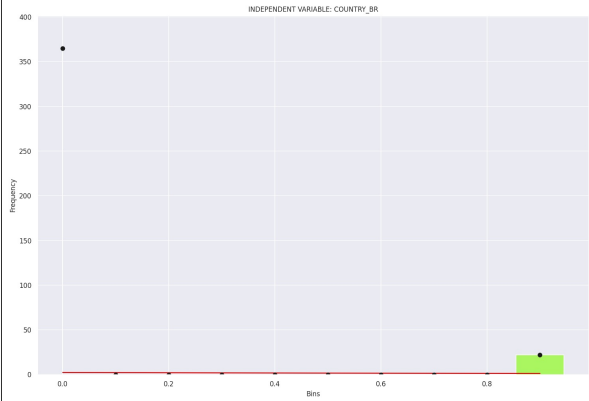
INDEPENDENT VARIABLE: CHANNEL_ONLINE



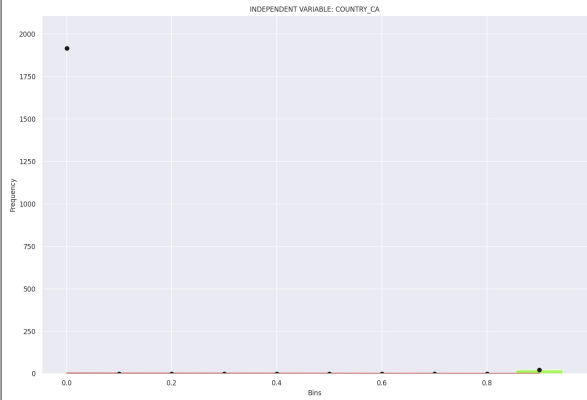
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1229	0	0	0	0	0
Share	63.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



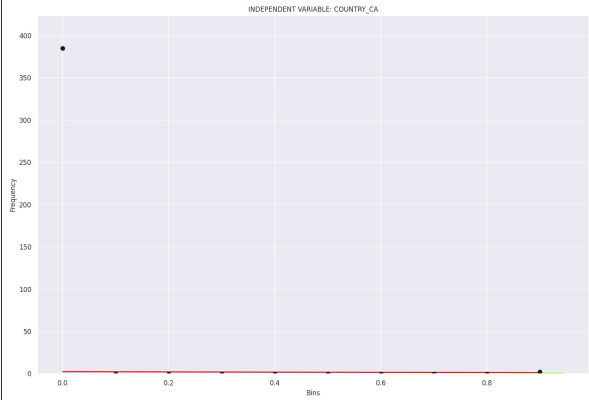
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	198	0	0	0	0	0
Share	51.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1894	0	0	0	0	0
Share	98.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

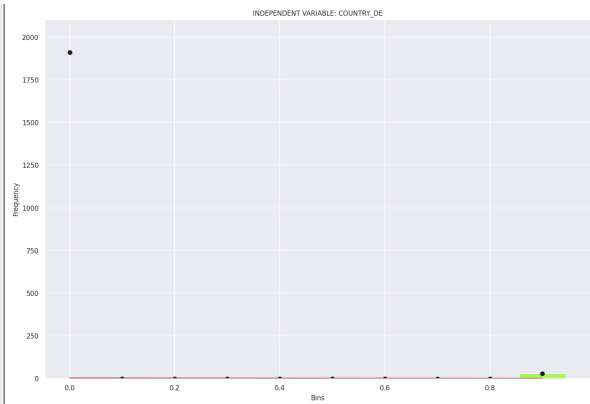


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	365	0	0	0	0	0
Share	94.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

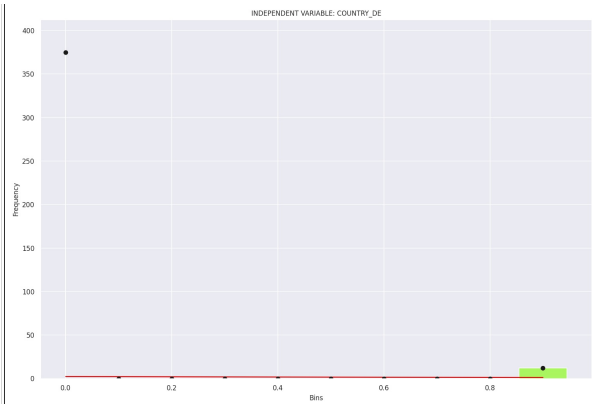


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1917	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

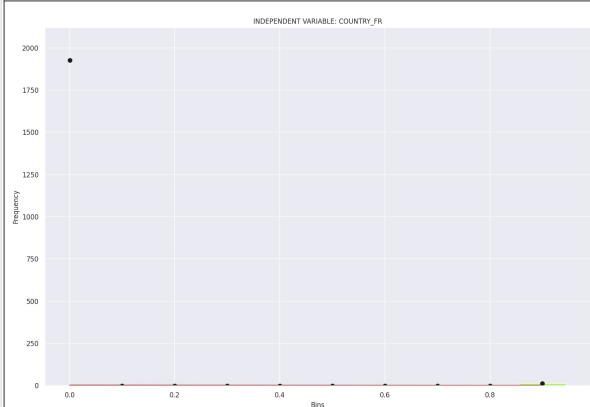
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	385	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



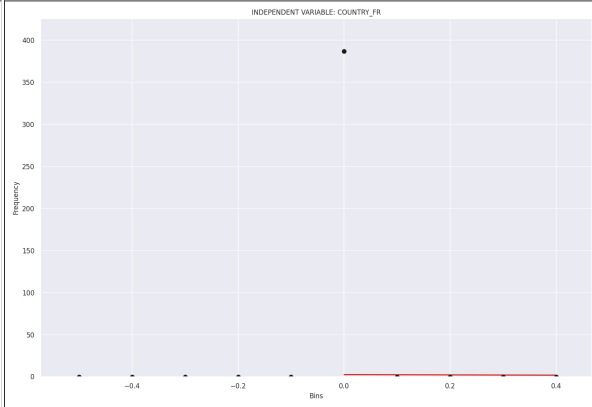
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1911	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



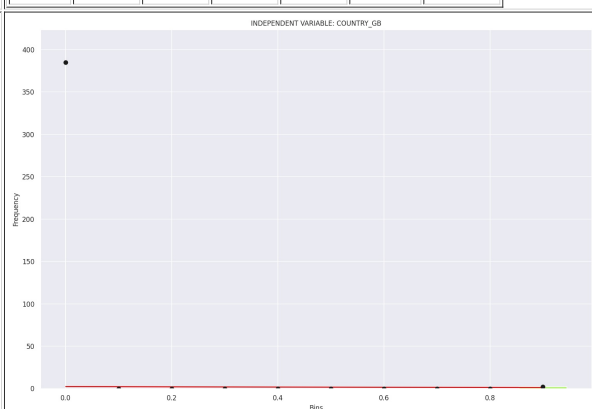
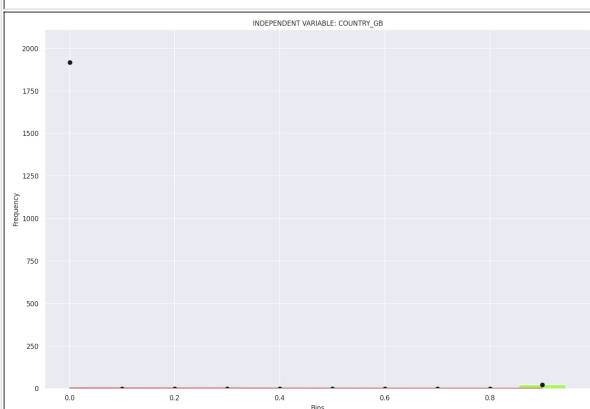
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	375	0	0	0	0	0
Share	97.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



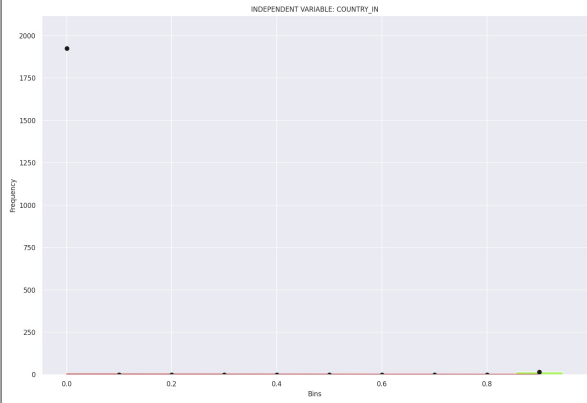
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1927	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



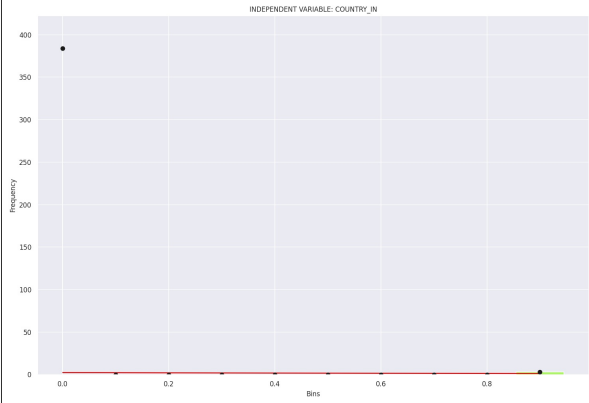
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



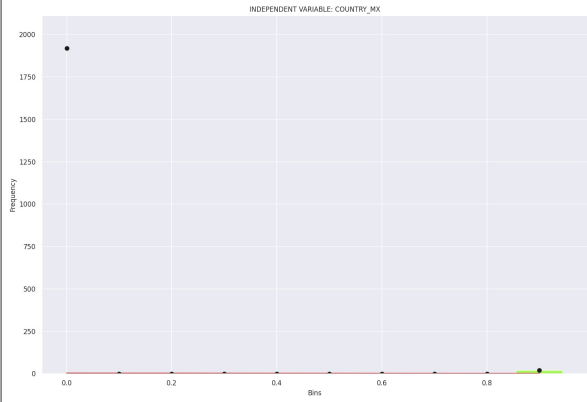
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1918	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



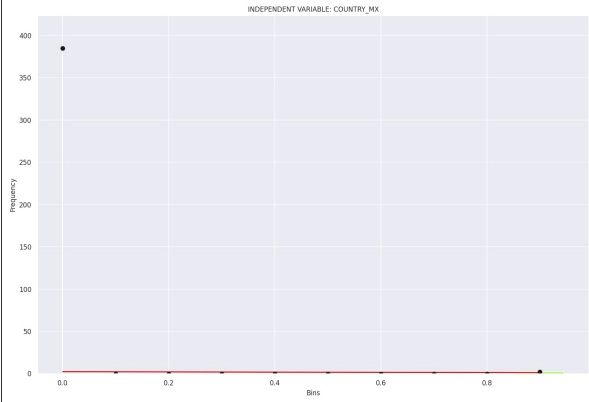
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	385	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1925	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

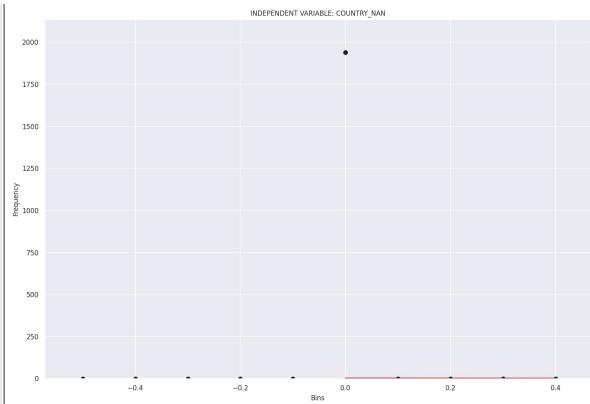


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	384	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

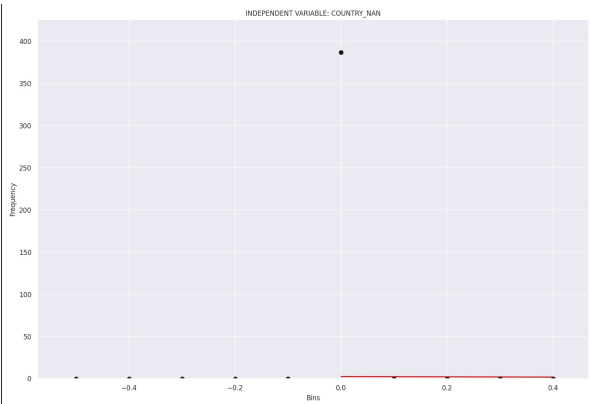


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1919	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

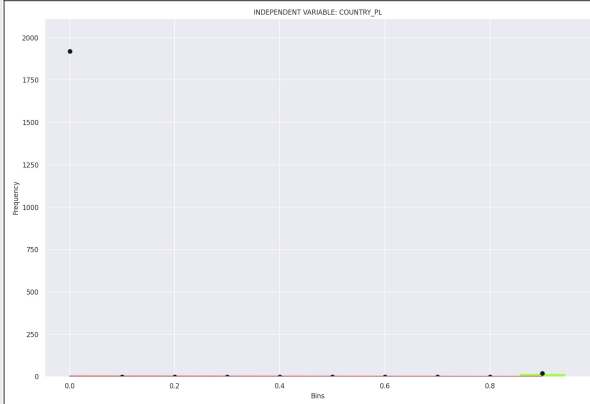
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	385	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



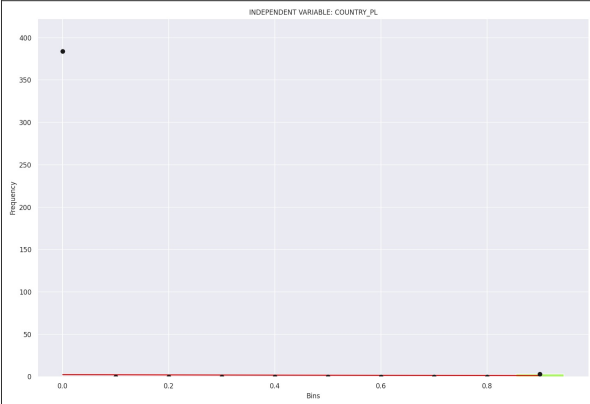
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



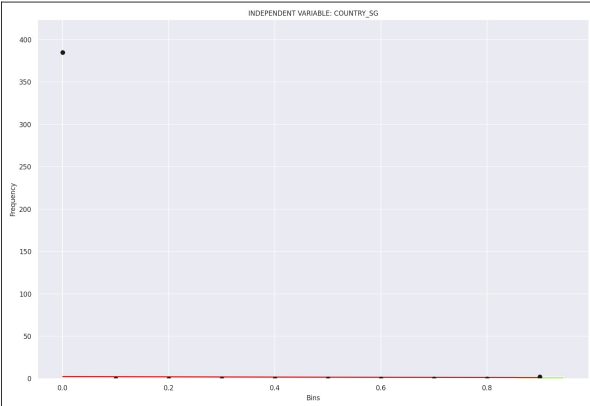
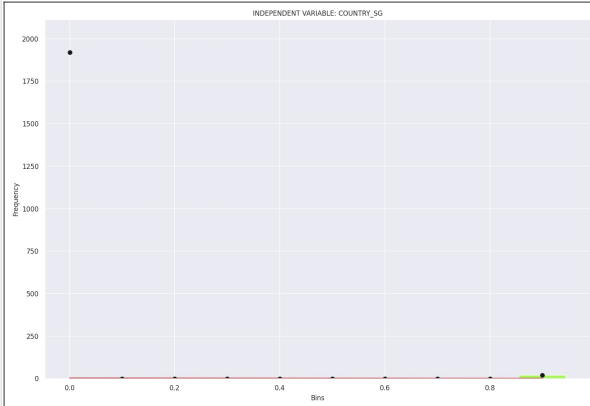
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



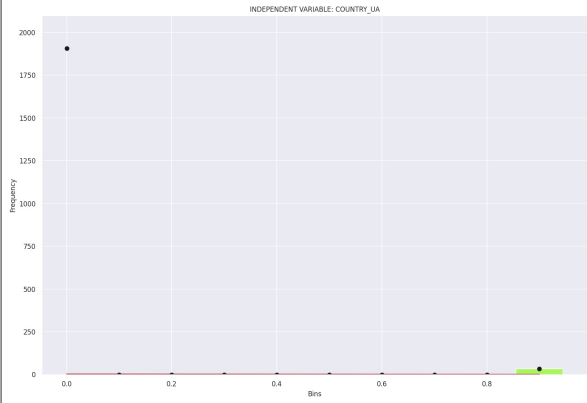
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1920	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



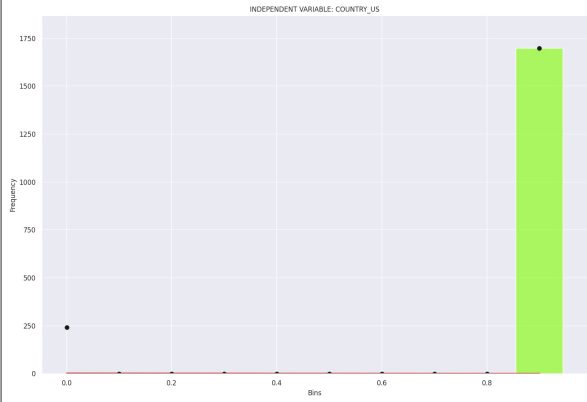
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	384	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1920	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

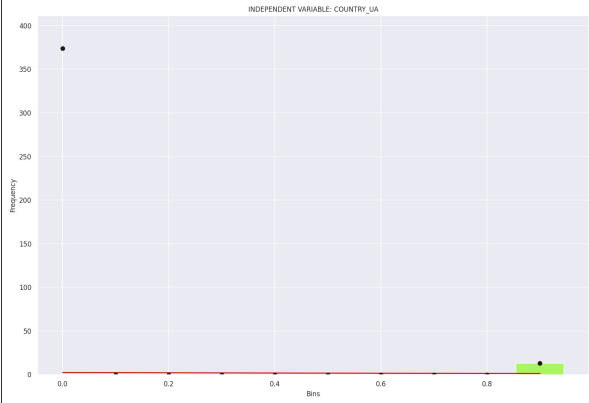


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1907	0	0	0	0	0
Share	98.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

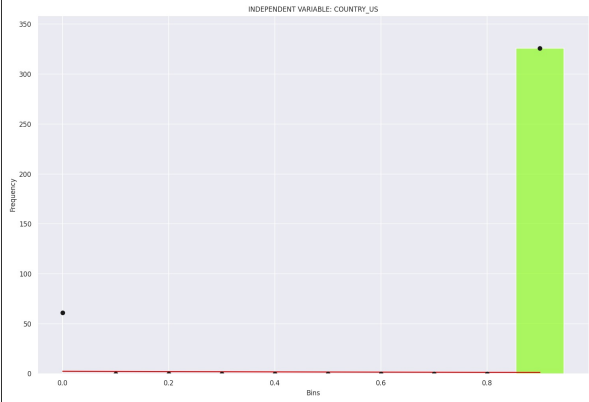


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	242	0	0	0	0	0
Share	12.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

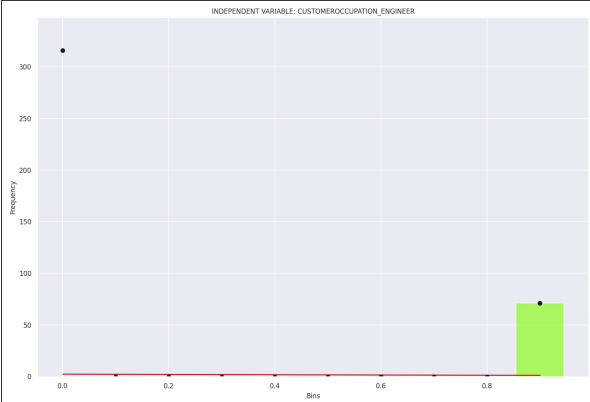
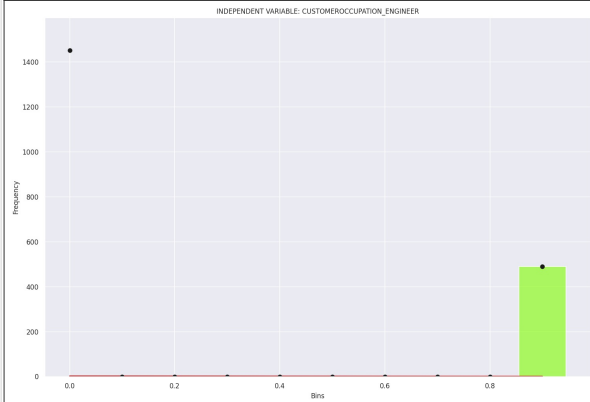
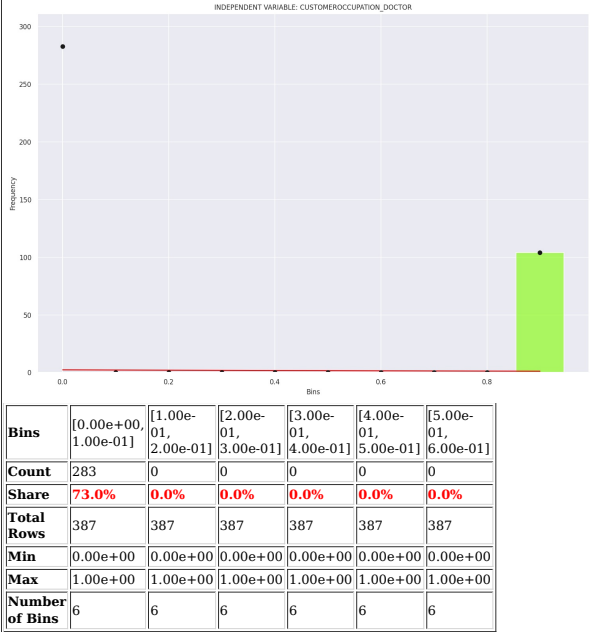
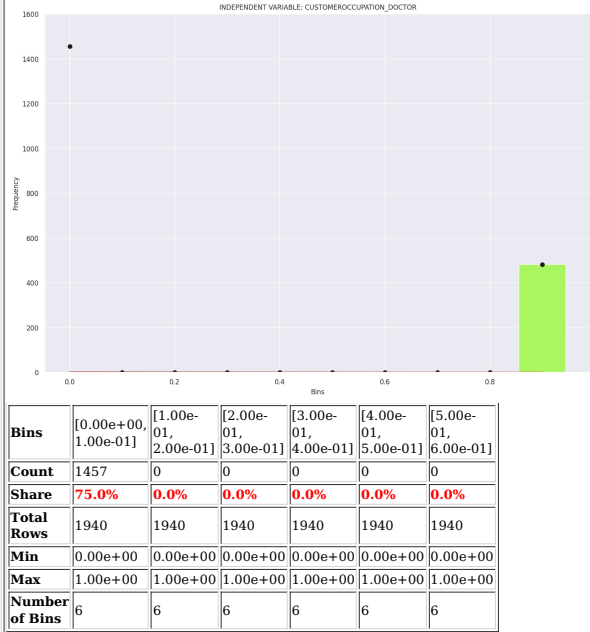
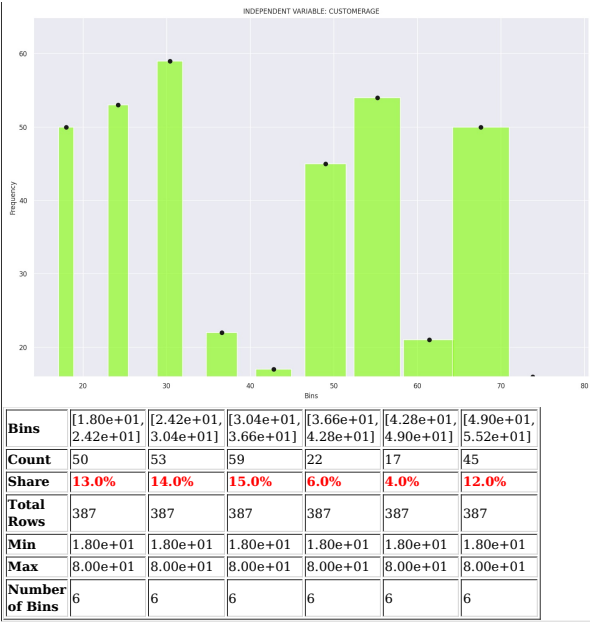
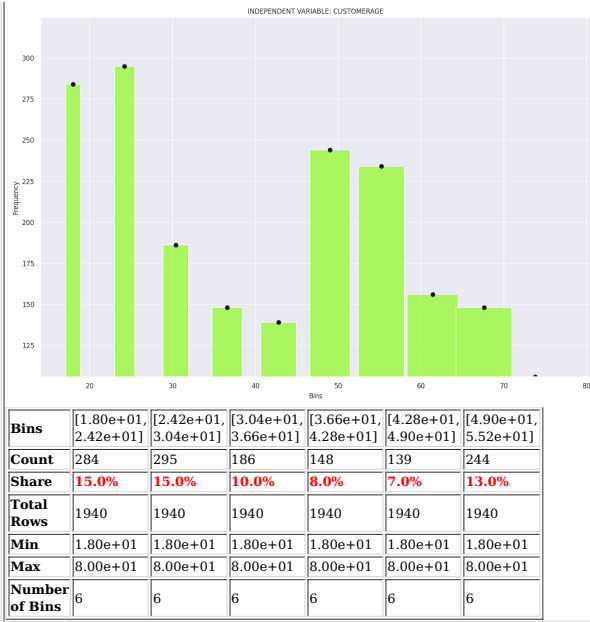
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	385	0	0	0	0	0
Share	99.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



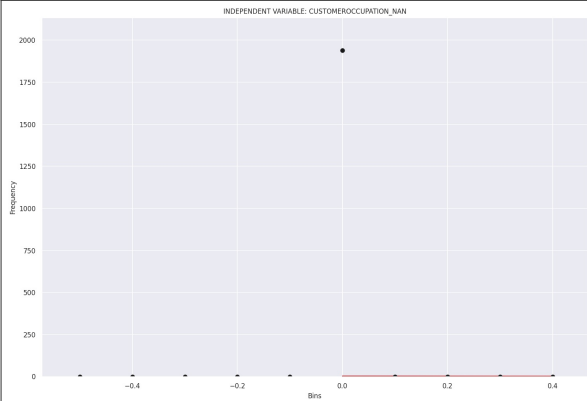
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	374	0	0	0	0	0
Share	97.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	61	0	0	0	0	0
Share	16.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

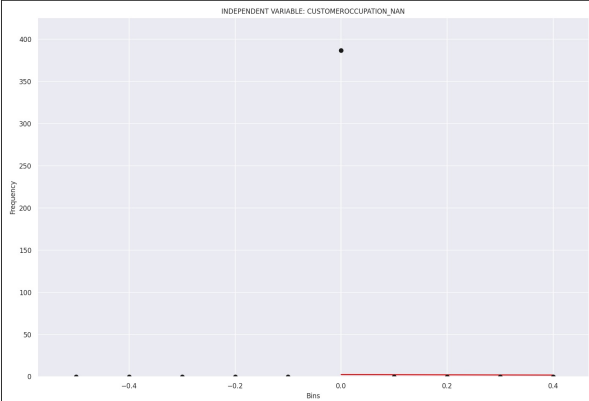


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1451	0	0	0	0	0
Share	75.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

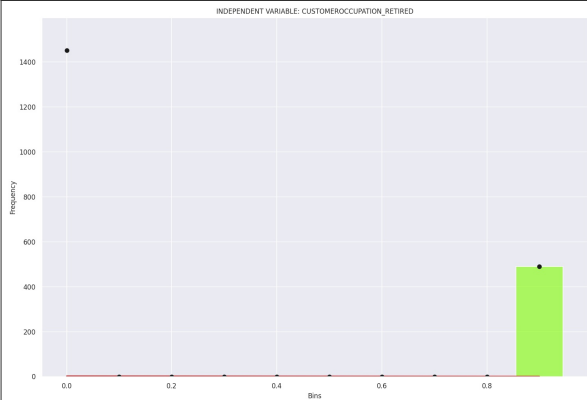


Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

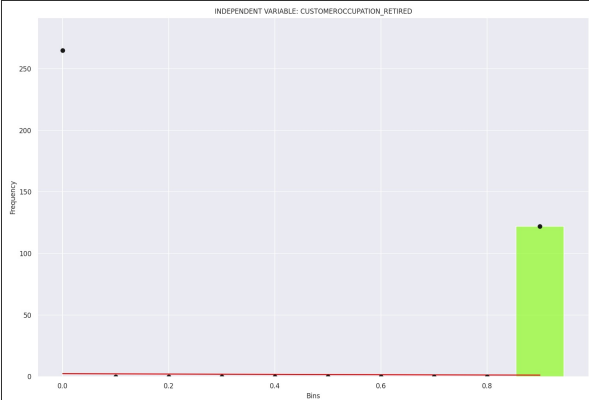
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	316	0	0	0	0	0
Share	82.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



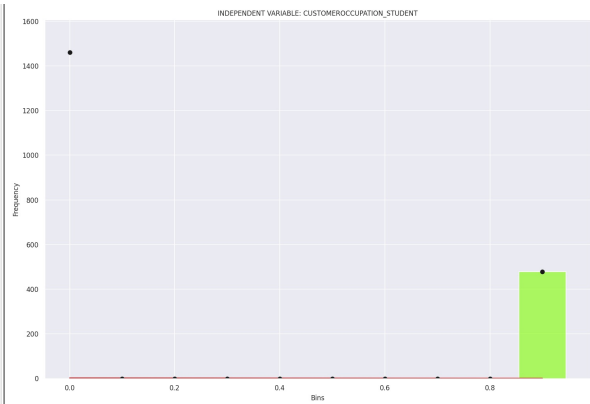
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



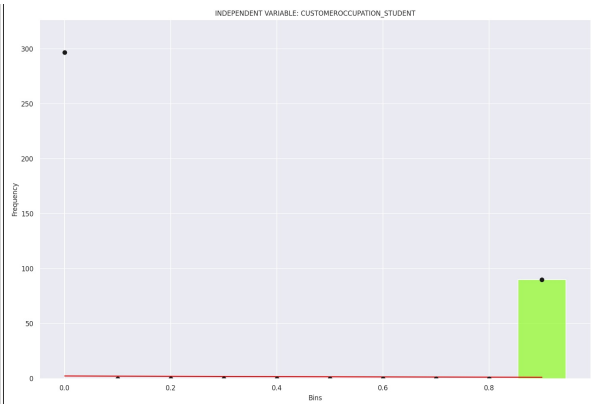
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1451	0	0	0	0	0
Share	75.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



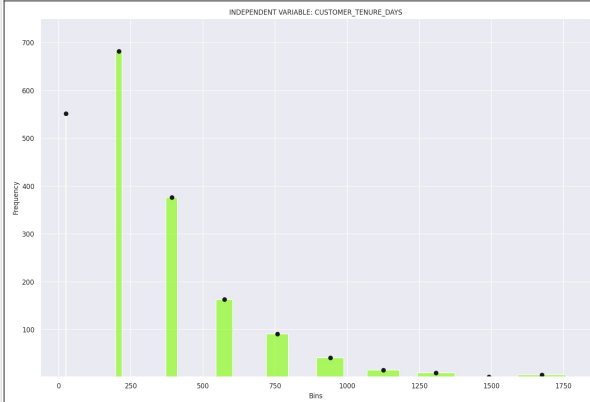
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	265	0	0	0	0	0
Share	68.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



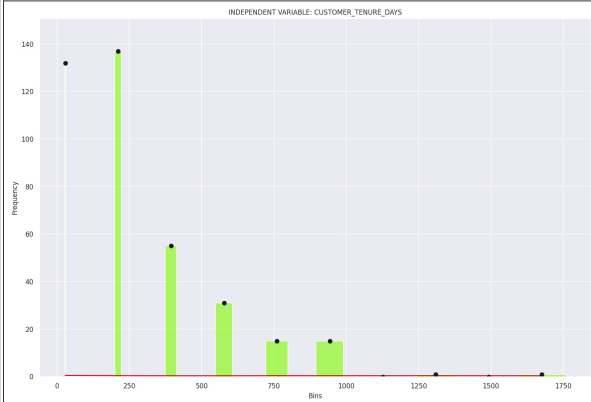
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1461	0	0	0	0	0
Share	75.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



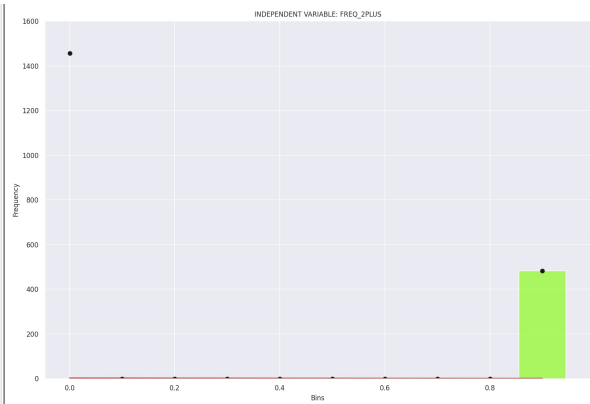
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	297	0	0	0	0	0
Share	77.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



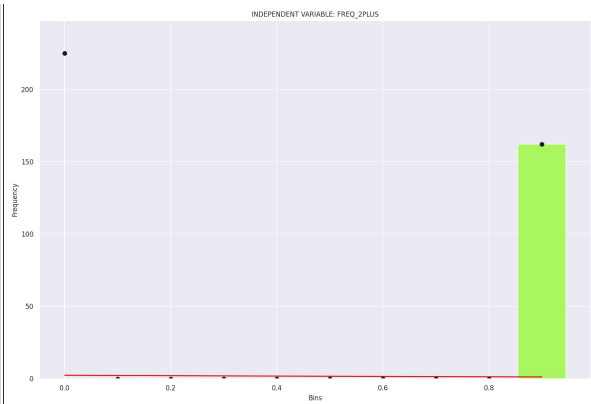
Bins	[2.50e+01, 2.08e+02]	[2.08e+02, 3.92e+02]	[3.92e+02, 5.75e+02]	[5.75e+02, 7.58e+02]	[7.58e+02, 9.42e+02]	[9.42e+02, 1.12e+03]
Count	552	682	377	163	91	41
Share	28.0%	35.0%	19.0%	8.0%	5.0%	2.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	2.50e+01	2.50e+01	2.50e+01	2.50e+01	2.50e+01	2.50e+01
Max	1.86e+03	1.86e+03	1.86e+03	1.86e+03	1.86e+03	1.86e+03
Number of Bins	6	6	6	6	6	6



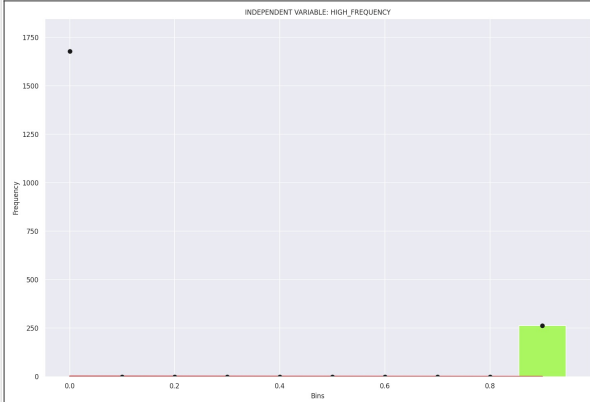
Bins	[2.84e+01, 2.11e+02]	[2.11e+02, 3.94e+02]	[3.94e+02, 5.77e+02]	[5.77e+02, 7.60e+02]	[7.60e+02, 9.43e+02]	[9.43e+02, 1.13e+03]
Count	132	137	55	31	15	15
Share	34.0%	35.0%	14.0%	8.0%	4.0%	4.0%
Total Rows	387	387	387	387	387	387
Min	2.80e+01	2.80e+01	2.80e+01	2.80e+01	2.80e+01	2.80e+01
Max	1.86e+03	1.86e+03	1.86e+03	1.86e+03	1.86e+03	1.86e+03
Number of Bins	6	6	6	6	6	6



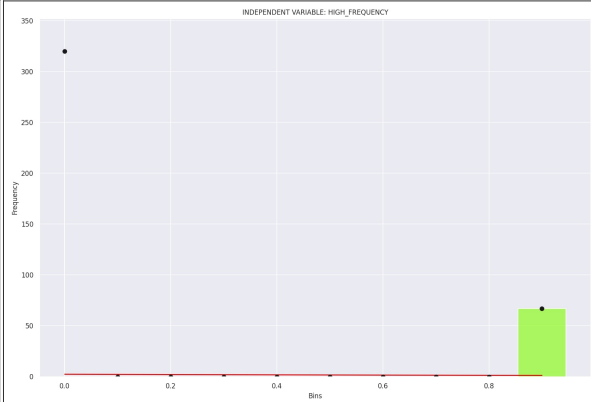
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1457	0	0	0	0	0
Share	75.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



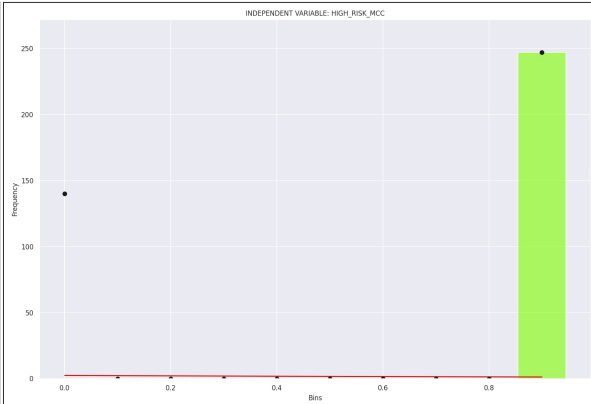
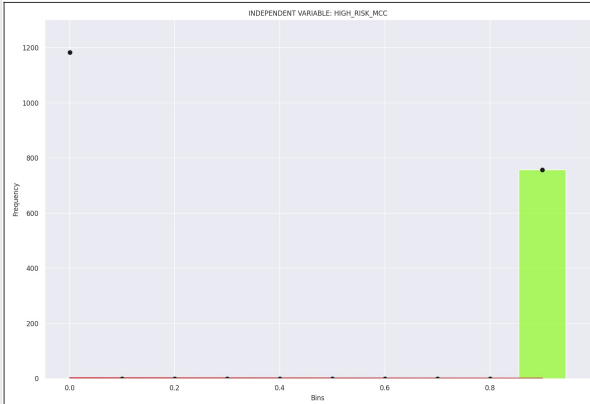
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	225	0	0	0	0	0
Share	58.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



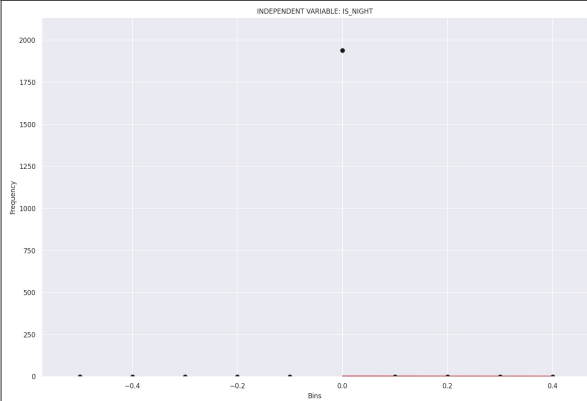
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1678	0	0	0	0	0
Share	86.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



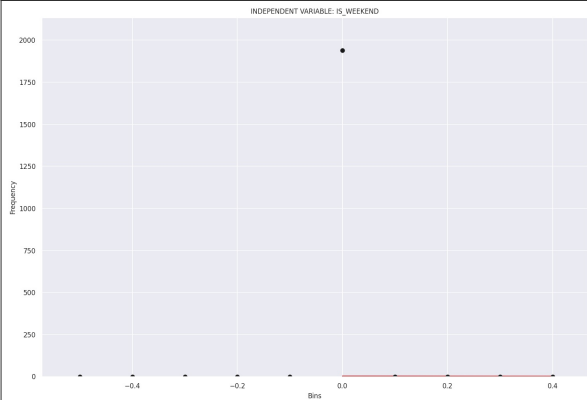
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	320	0	0	0	0	0
Share	83.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1183	0	0	0	0	0
Share	61.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

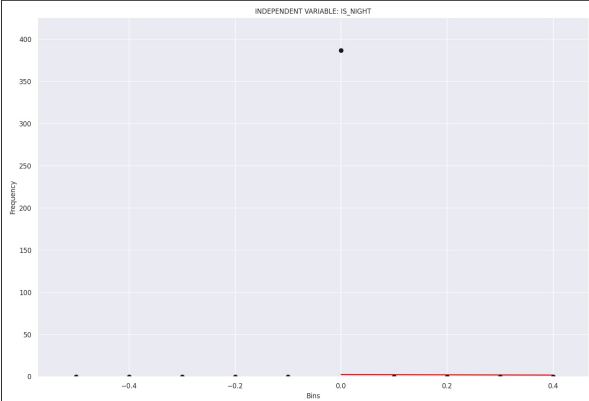


Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

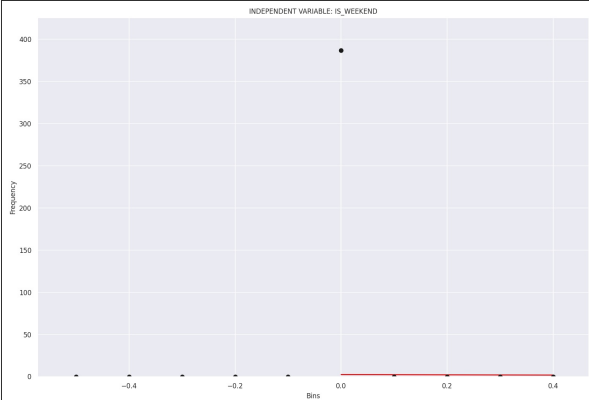


Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

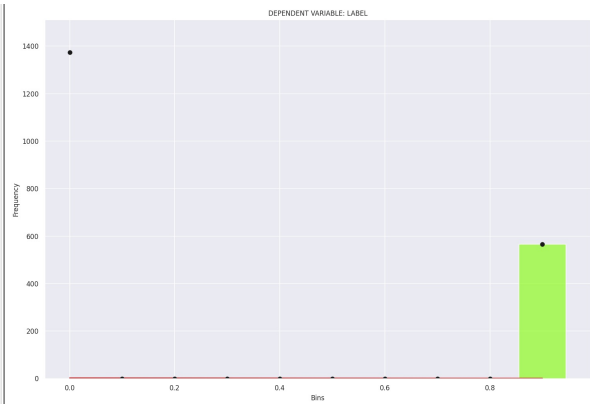
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	140	0	0	0	0	0
Share	36.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



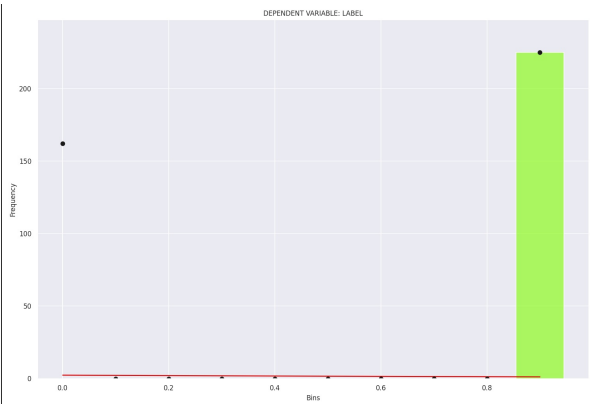
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



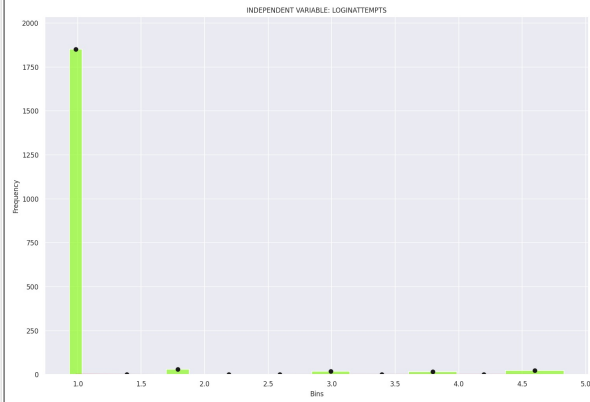
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



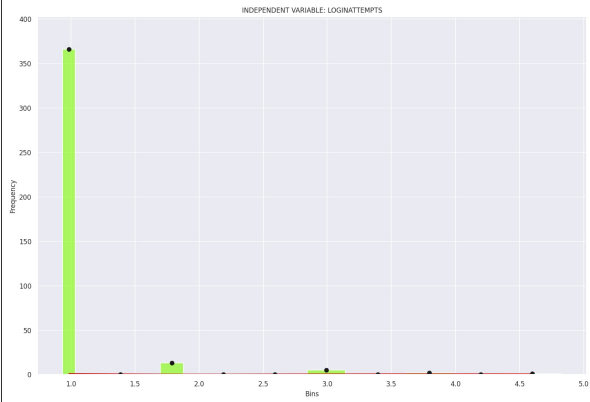
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1374	0	0	0	0	0
Share	71.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



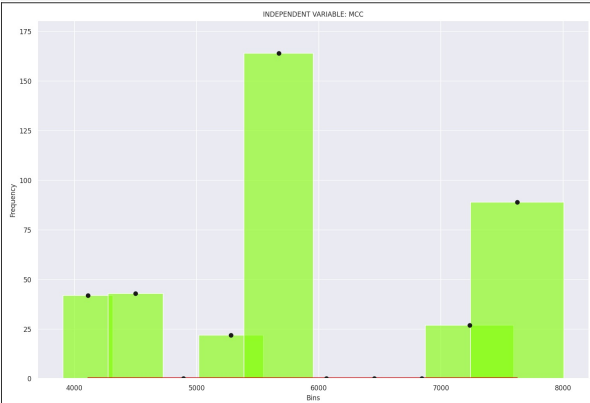
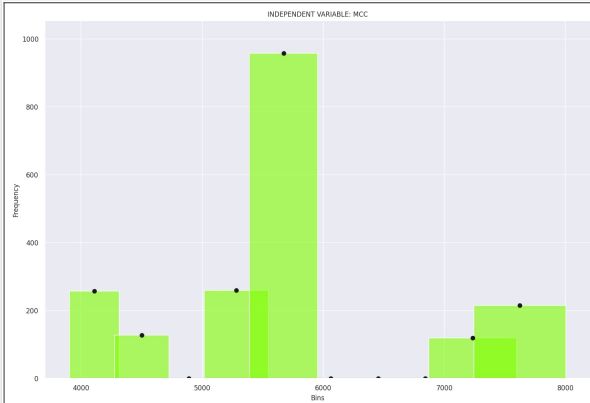
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	162	0	0	0	0	0
Share	42.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



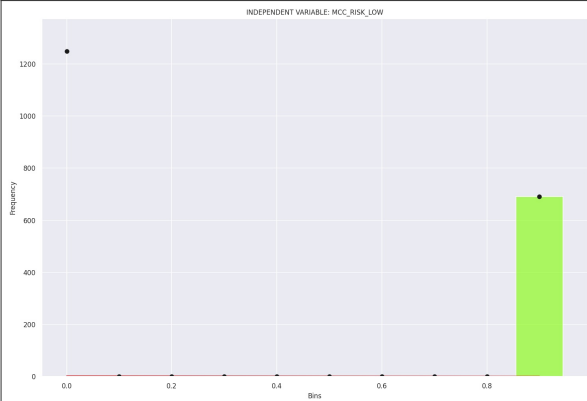
Bins	[9.85e-01, 1.39e+00]	[1.39e+00, 1.79e+00]	[1.79e+00, 2.19e+00]	[2.19e+00, 2.59e+00]	[2.59e+00, 2.99e+00]	[2.99e+00, 3.39e+00]
Count	1851	0	30	0	0	20
Share	95.0%	0.0%	2.0%	0.0%	0.0%	1.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00
Number of Bins	6	6	6	6	6	6



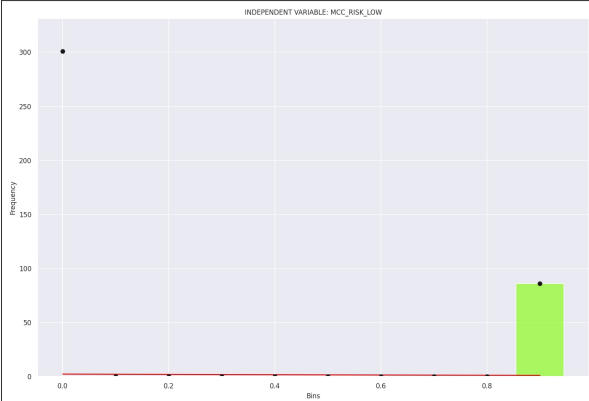
Bins	[9.85e-01, 1.39e+00]	[1.39e+00, 1.79e+00]	[1.79e+00, 2.19e+00]	[2.19e+00, 2.59e+00]	[2.59e+00, 2.99e+00]	[2.99e+00, 3.39e+00]
Count	366	0	13	0	0	5
Share	95.0%	0.0%	3.0%	0.0%	0.0%	1.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00
Number of Bins	6	6	6	6	6	6



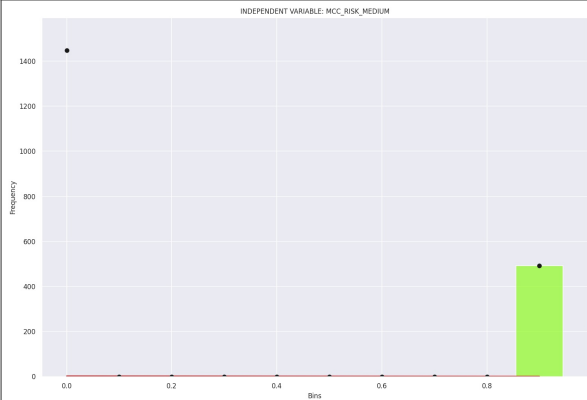
Bins	[4.11e+03, 4.50e+03]	[4.50e+03, 4.89e+03]	[4.89e+03, 5.28e+03]	[5.28e+03, 5.67e+03]	[5.67e+03, 6.06e+03]	[6.06e+03, 6.45e+03]
Count	258	128	0	260	958	0
Share	13.0%	7.0%	0.0%	13.0%	49.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	4.11e+03	4.11e+03	4.11e+03	4.11e+03	4.11e+03	4.11e+03
Max	8.01e+03	8.01e+03	8.01e+03	8.01e+03	8.01e+03	8.01e+03
Number of Bins	6	6	6	6	6	6



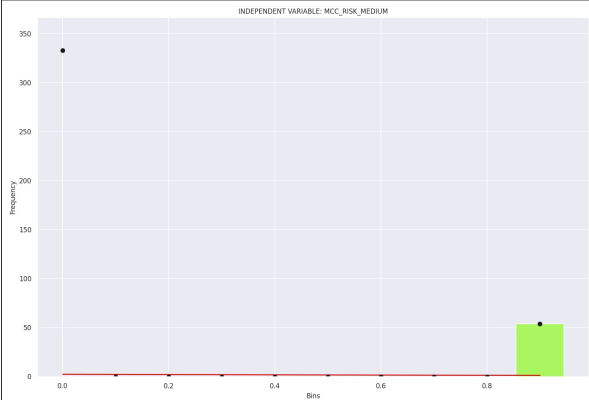
Bins	[4.11e+03, 4.50e+03]	[4.50e+03, 4.89e+03]	[4.89e+03, 5.28e+03]	[5.28e+03, 5.67e+03]	[5.67e+03, 6.06e+03]	[6.06e+03, 6.45e+03]
Count	42	43	0	22	164	0
Share	11.0%	11.0%	0.0%	6.0%	42.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	4.11e+03	4.11e+03	4.11e+03	4.11e+03	4.11e+03	4.11e+03
Max	8.01e+03	8.01e+03	8.01e+03	8.01e+03	8.01e+03	8.01e+03
Number of Bins	6	6	6	6	6	6



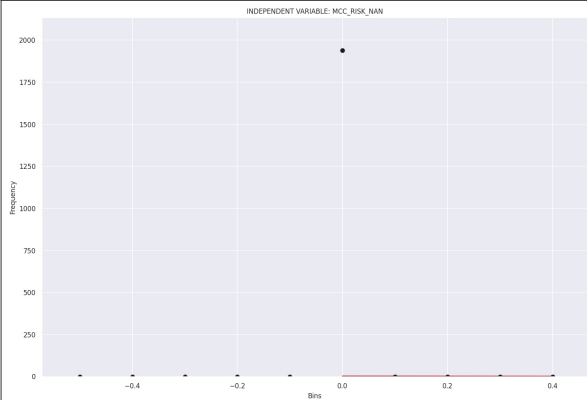
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1249	0	0	0	0	0
Share	64.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



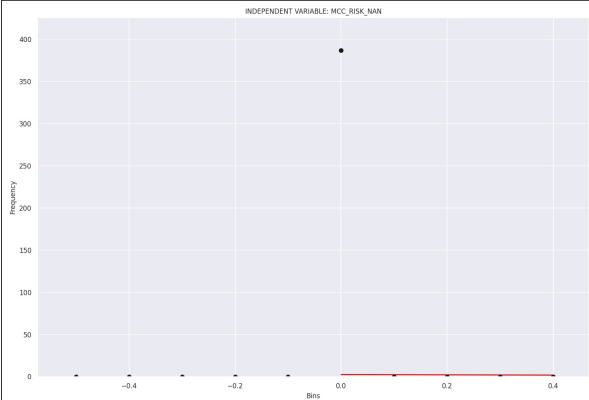
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	301	0	0	0	0	0
Share	78.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



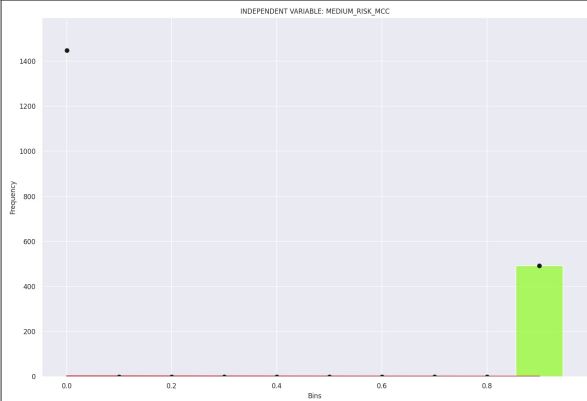
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1448	0	0	0	0	0
Share	75.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	333	0	0	0	0	0
Share	86.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

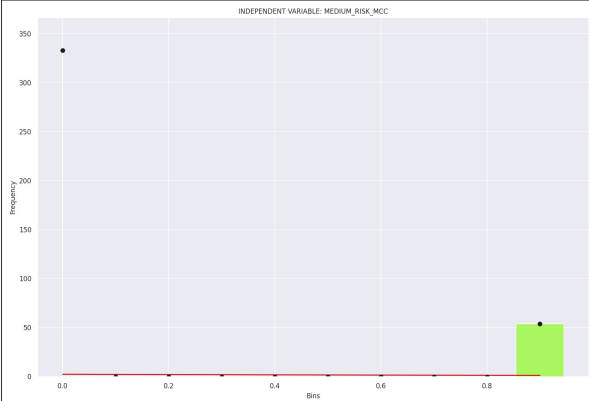


Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

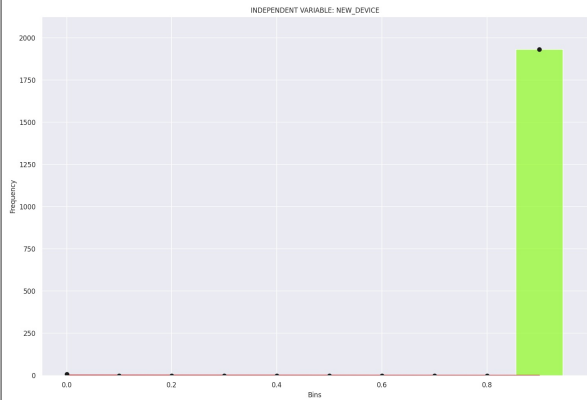


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1448	0	0	0	0	0
Share	75.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

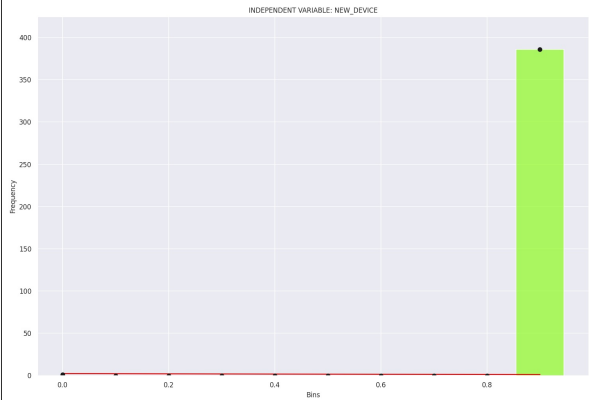
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6



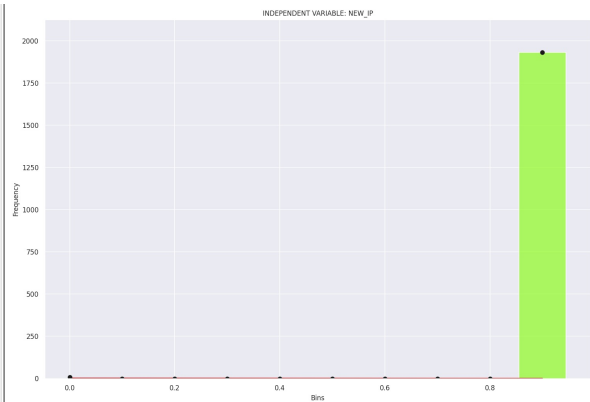
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	333	0	0	0	0	0
Share	86.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



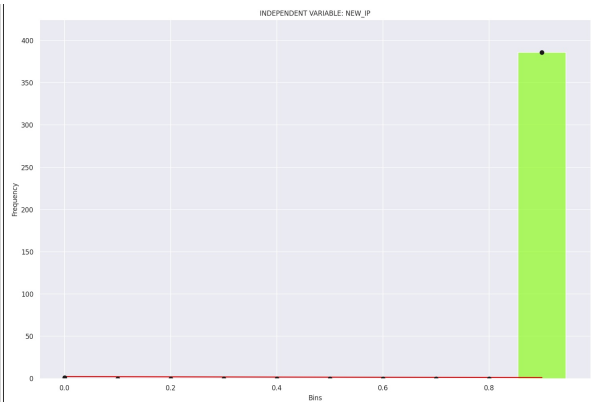
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	8	0	0	0	0	0
Share	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



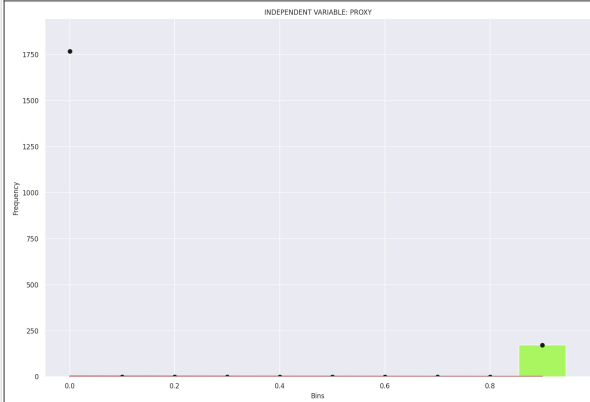
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1	0	0	0	0	0
Share	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



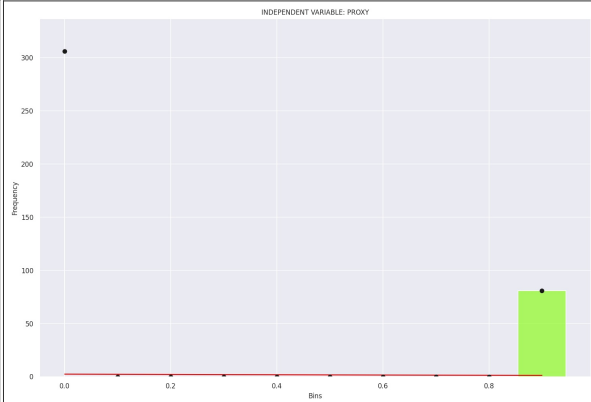
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	8	0	0	0	0	0
Share	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



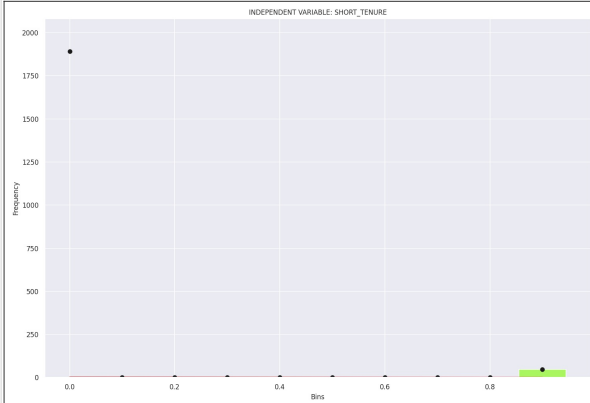
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1	0	0	0	0	0
Share	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



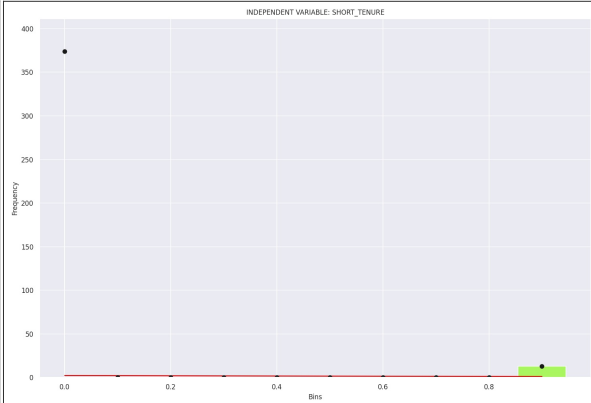
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1768	0	0	0	0	0
Share	91.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



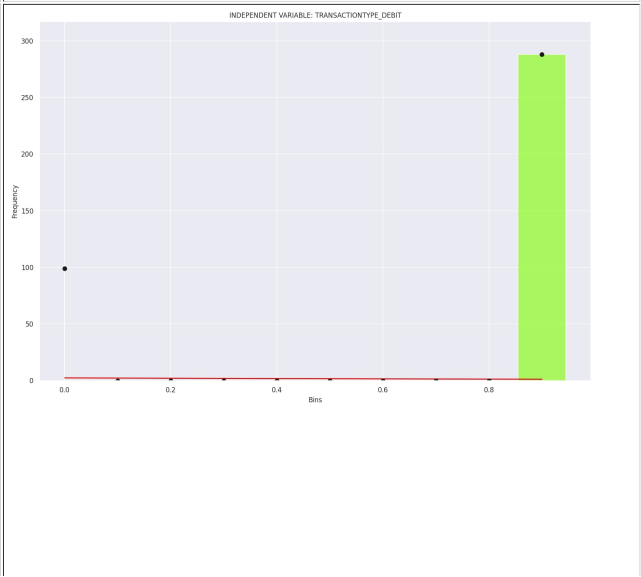
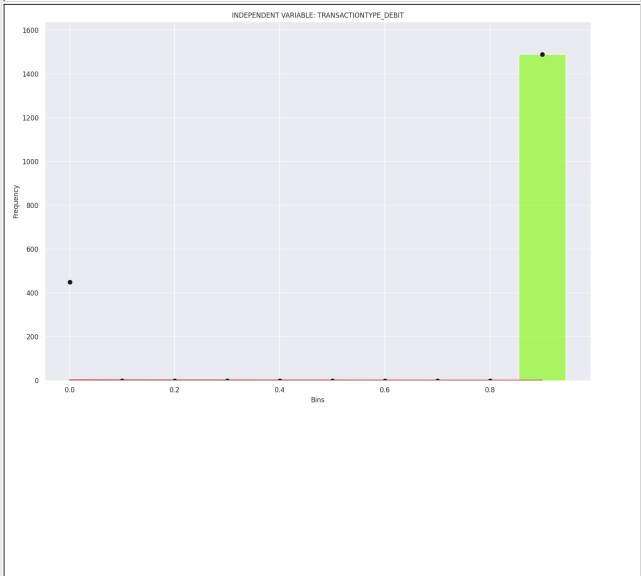
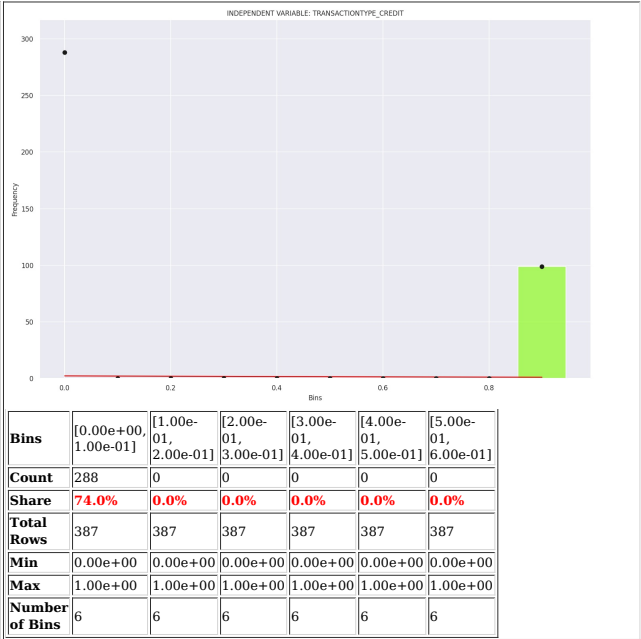
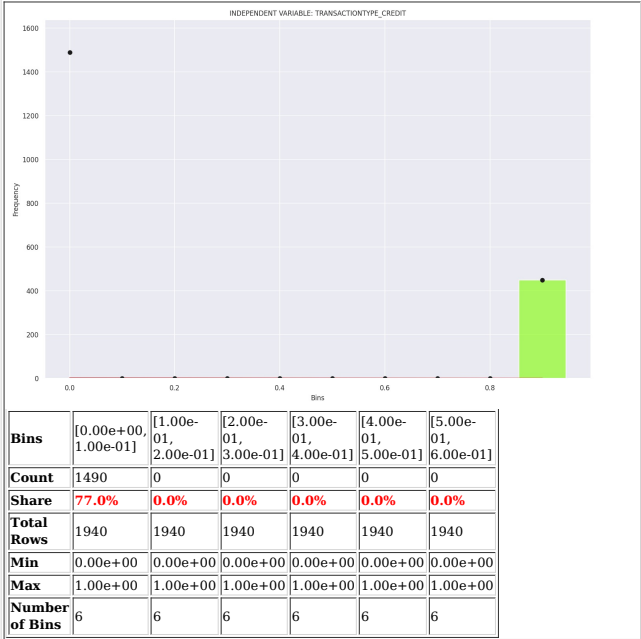
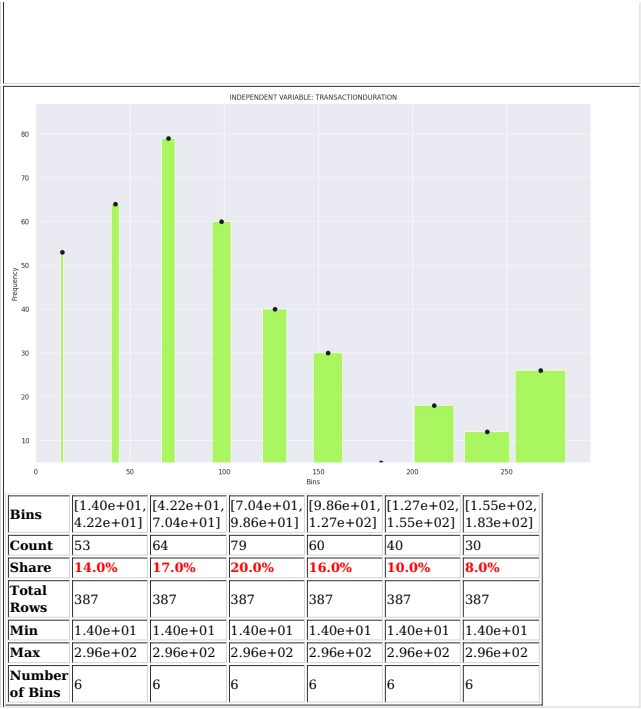
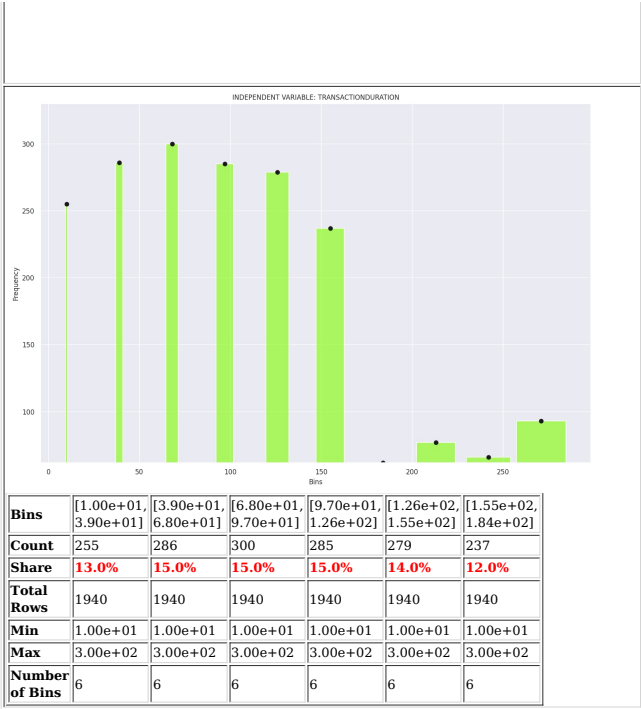
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	306	0	0	0	0	0
Share	79.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



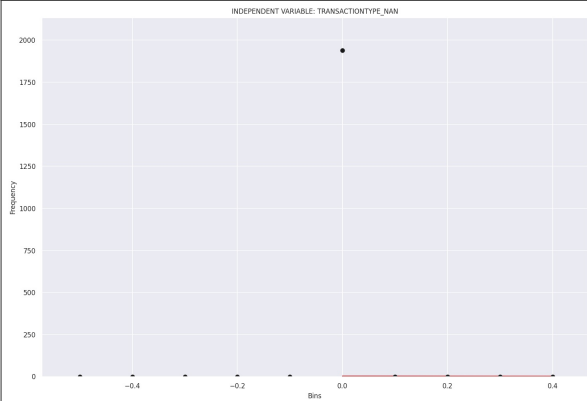
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1892	0	0	0	0	0
Share	98.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



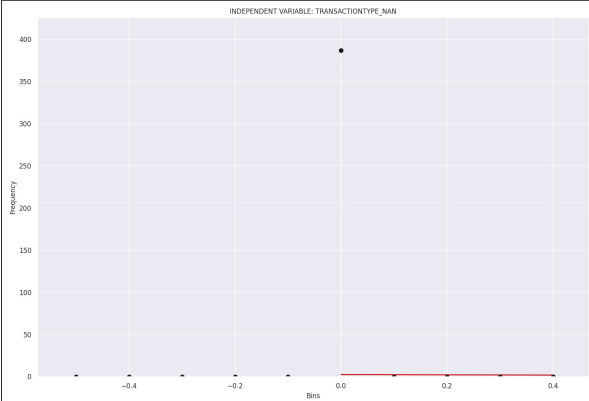
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	374	0	0	0	0	0
Share	97.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	450	0	0	0	0	0
Share	23.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

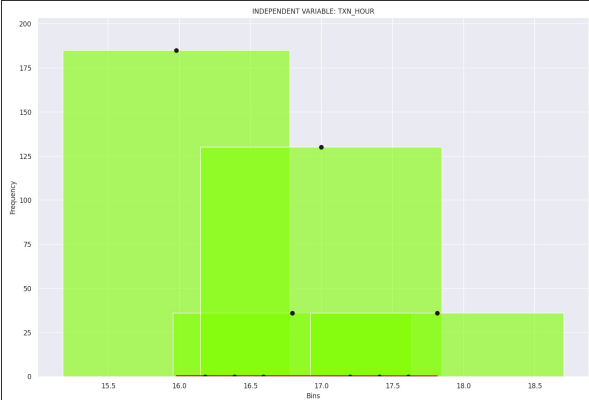
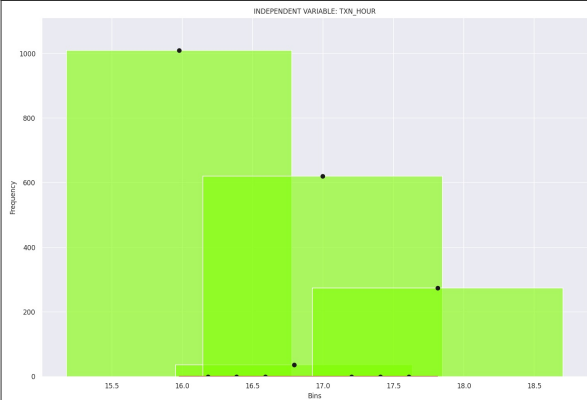


Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	99	0	0	0	0	0
Share	26.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6



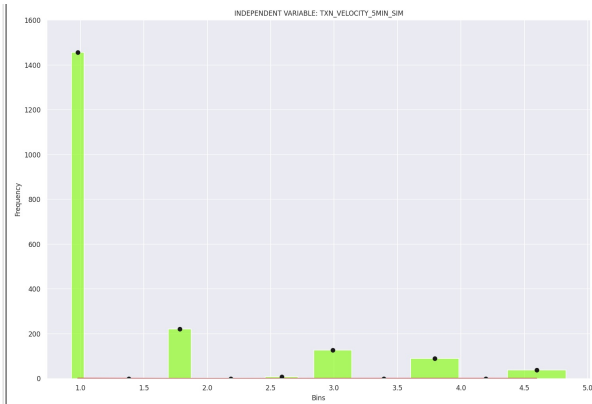
Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	1940
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

Bins	[-5.00e-01, -4.00e-01]	[-4.00e-01, -3.00e-01]	[-3.00e-01, -2.00e-01]	[-2.00e-01, -1.00e-01]	[-1.00e-01, 0.00e+00]	[0.00e+00, 1.00e-01]
Count	0	0	0	0	0	387
Share	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Number of Bins	6	6	6	6	6	6

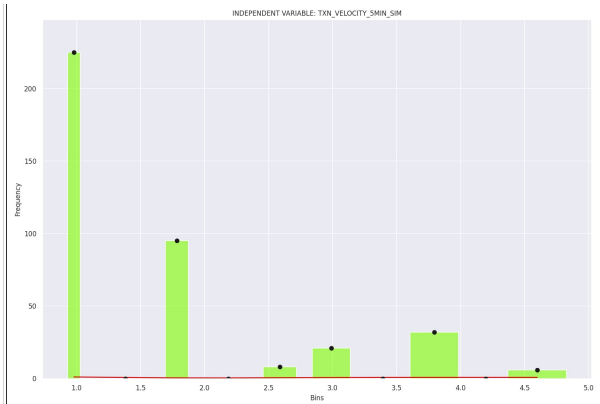


Bins	[1.60e+01, 1.62e+01]	[1.62e+01, 1.64e+01]	[1.64e+01, 1.66e+01]	[1.66e+01, 1.68e+01]	[1.68e+01, 1.70e+01]	[1.70e+01, 1.72e+01]
Count	1010	0	0	0	36	620
Share	52.0%	0.0%	0.0%	0.0%	2.0%	32.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	1.50e+01	1.50e+01	1.50e+01	1.50e+01	1.50e+01	1.50e+01
Max	1.80e+01	1.80e+01	1.80e+01	1.80e+01	1.80e+01	1.80e+01
Number of Bins	6	6	6	6	6	6

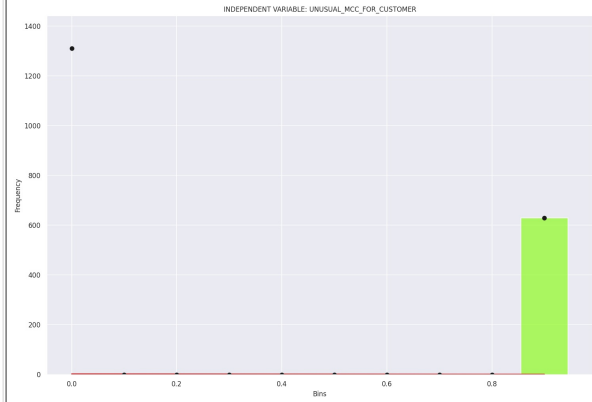
Bins	[1.60e+01, 1.62e+01]	[1.62e+01, 1.64e+01]	[1.64e+01, 1.66e+01]	[1.66e+01, 1.68e+01]	[1.68e+01, 1.70e+01]	[1.70e+01, 1.72e+01]
Count	185	0	0	0	36	130
Share	48.0%	0.0%	0.0%	0.0%	9.0%	34.0%
Total Rows	387	387	387	387	387	387
Min	1.50e+01	1.50e+01	1.50e+01	1.50e+01	1.50e+01	1.50e+01
Max	1.80e+01	1.80e+01	1.80e+01	1.80e+01	1.80e+01	1.80e+01
Number of Bins	6	6	6	6	6	6



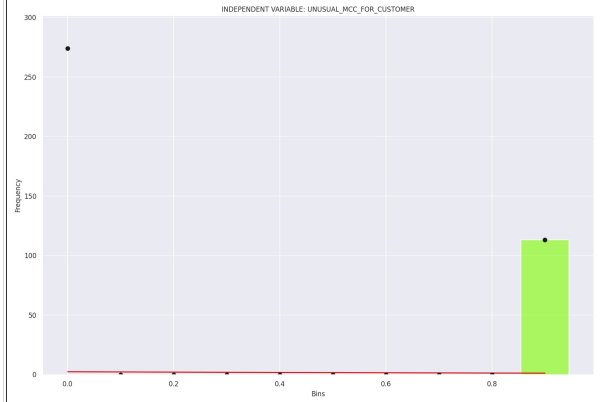
Bins	[9.79e-01, 1.38e+00]	[1.38e+00, 1.78e+00]	[1.78e+00, 2.19e+00]	[2.19e+00, 2.59e+00]	[2.59e+00, 2.99e+00]	[2.99e+00, 3.39e+00]
Count	1457	0	221	0	8	127
Share	75.0%	0.0%	11.0%	0.0%	0.0%	7.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00
Number of Bins	6	6	6	6	6	6



Bins	[9.79e-01, 1.38e+00]	[1.38e+00, 1.78e+00]	[1.78e+00, 2.19e+00]	[2.19e+00, 2.59e+00]	[2.59e+00, 2.99e+00]	[2.99e+00, 3.39e+00]
Count	225	0	95	0	8	21
Share	58.0%	0.0%	25.0%	0.0%	2.0%	5.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00	5.00e+00
Number of Bins	6	6	6	6	6	6

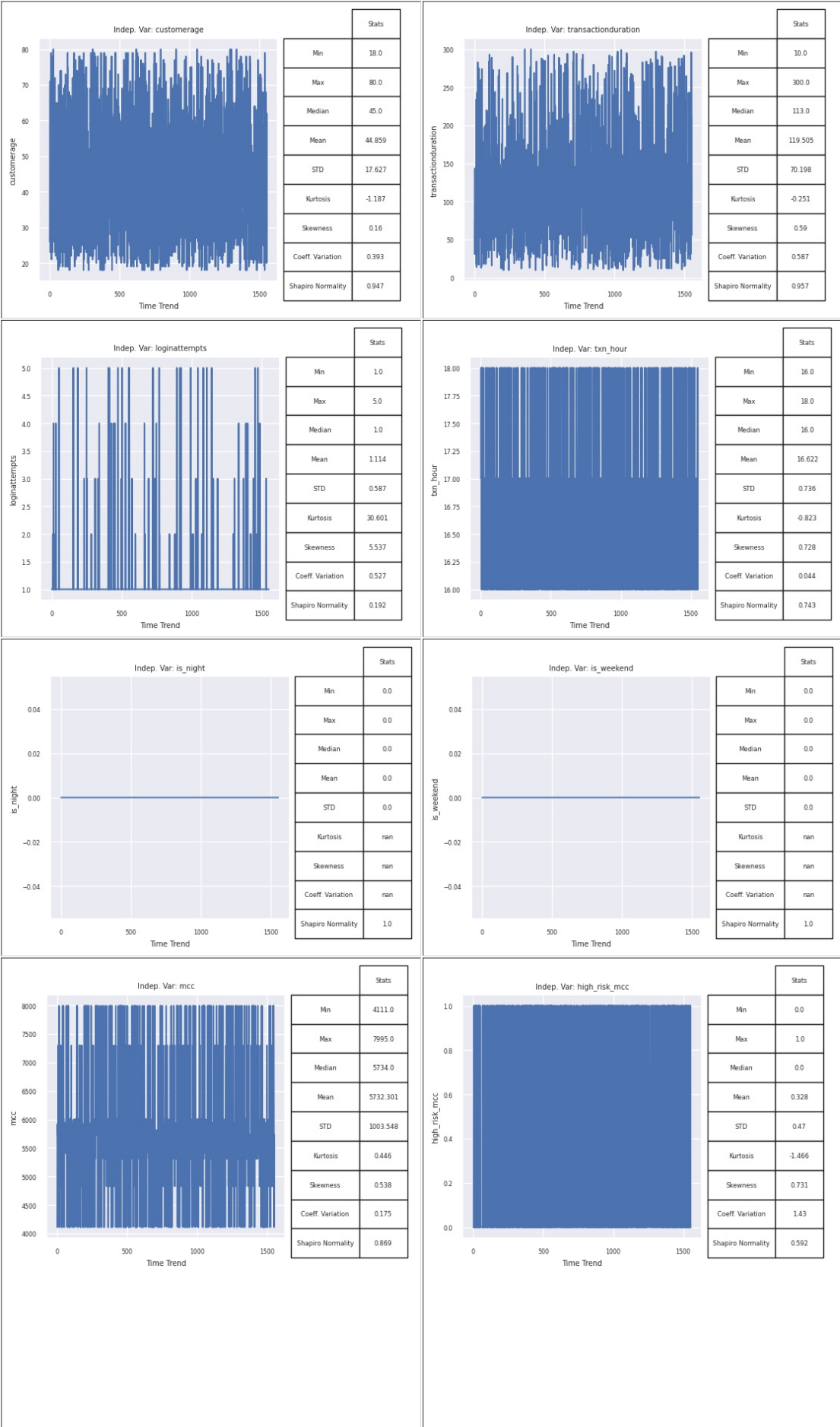


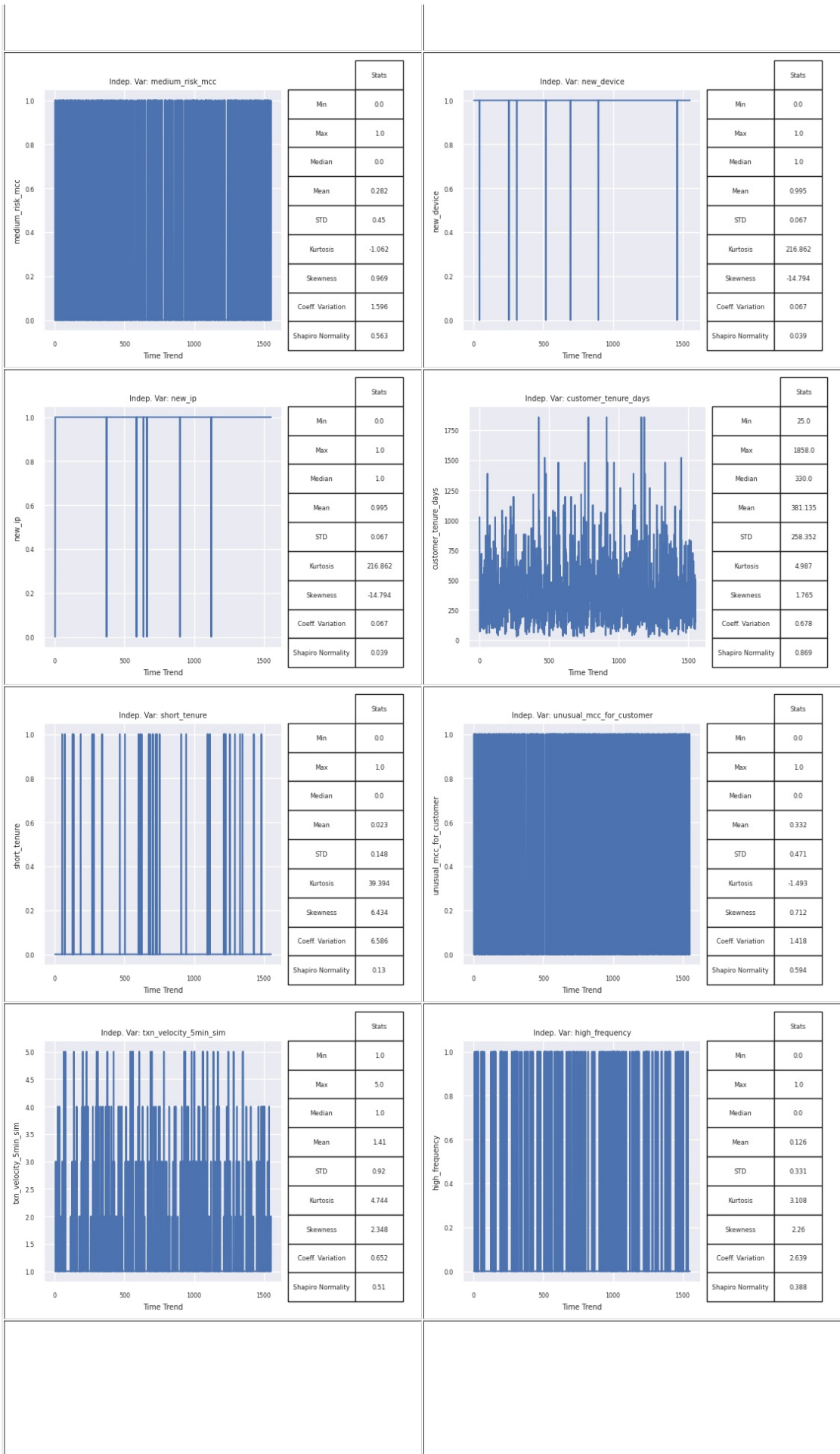
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	1311	0	0	0	0	0
Share	68.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	1940	1940	1940	1940	1940	1940
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

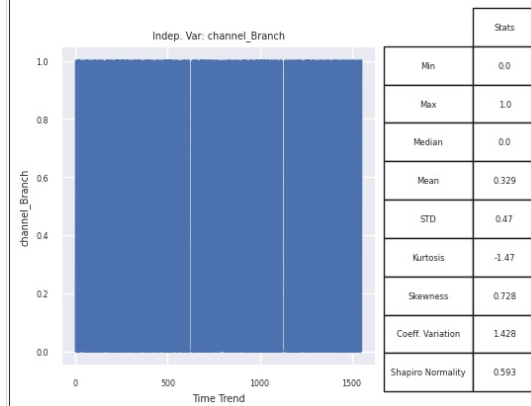
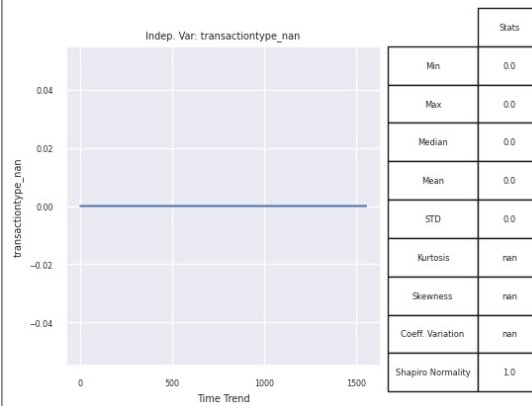
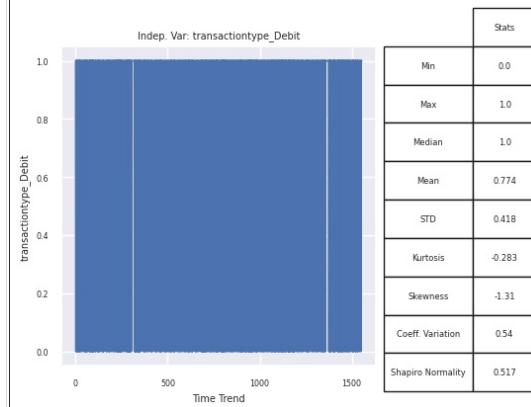
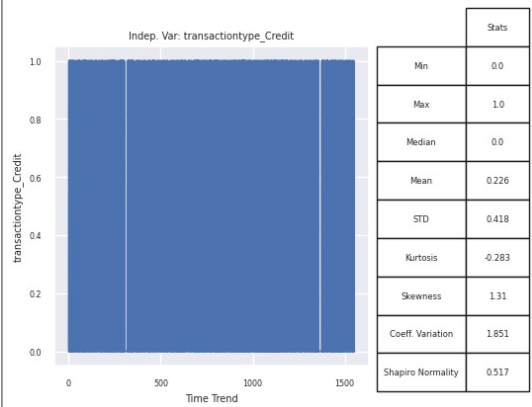
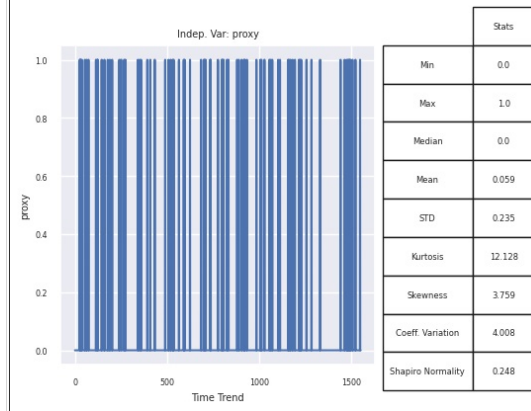
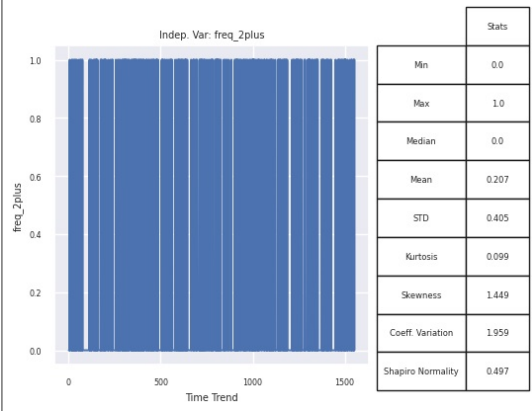


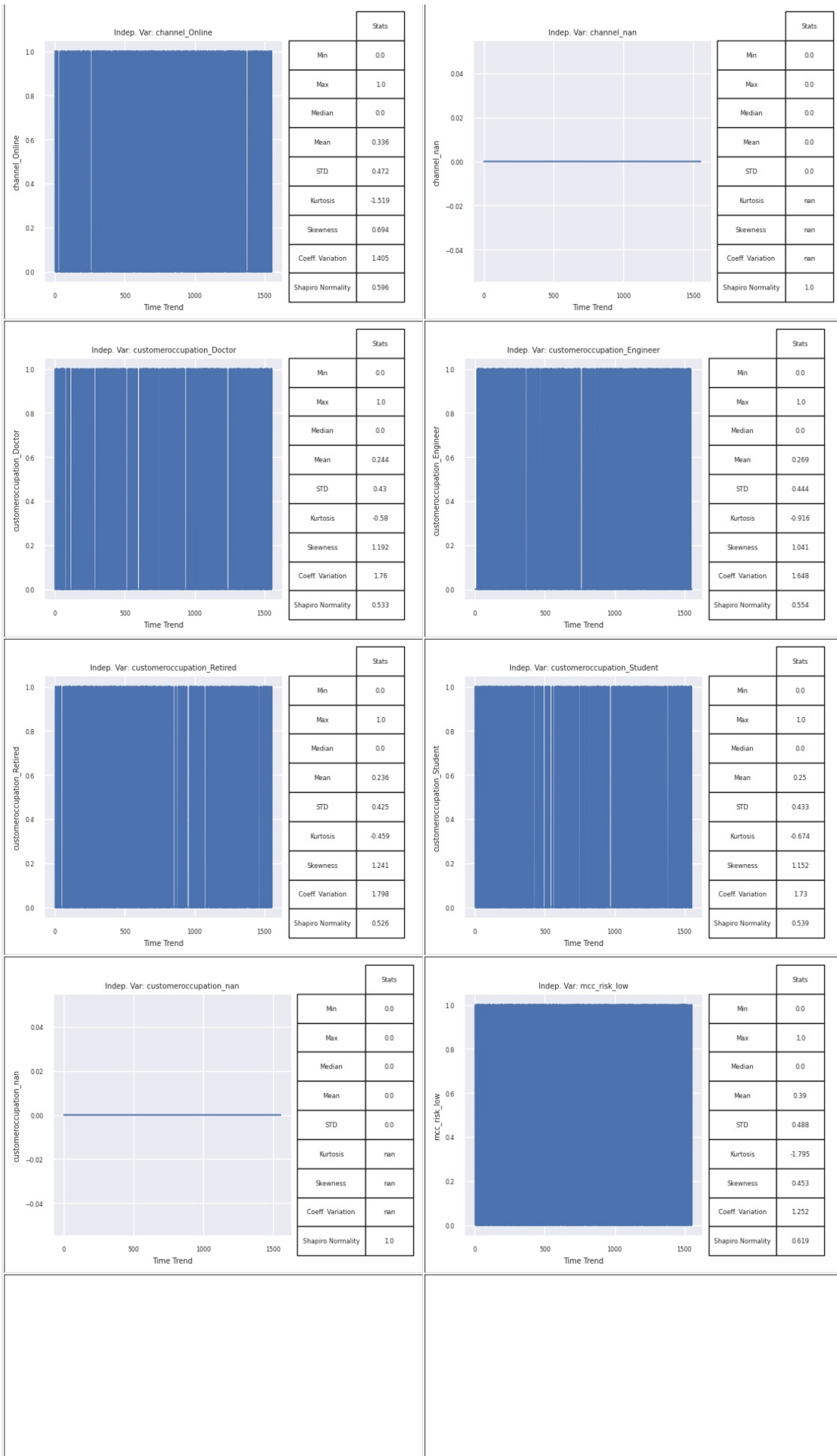
Bins	[0.00e+00, 1.00e-01]	[1.00e-01, 2.00e-01]	[2.00e-01, 3.00e-01]	[3.00e-01, 4.00e-01]	[4.00e-01, 5.00e-01]	[5.00e-01, 6.00e-01]
Count	274	0	0	0	0	0
Share	71.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total Rows	387	387	387	387	387	387
Min	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Max	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00	1.00e+00
Number of Bins	6	6	6	6	6	6

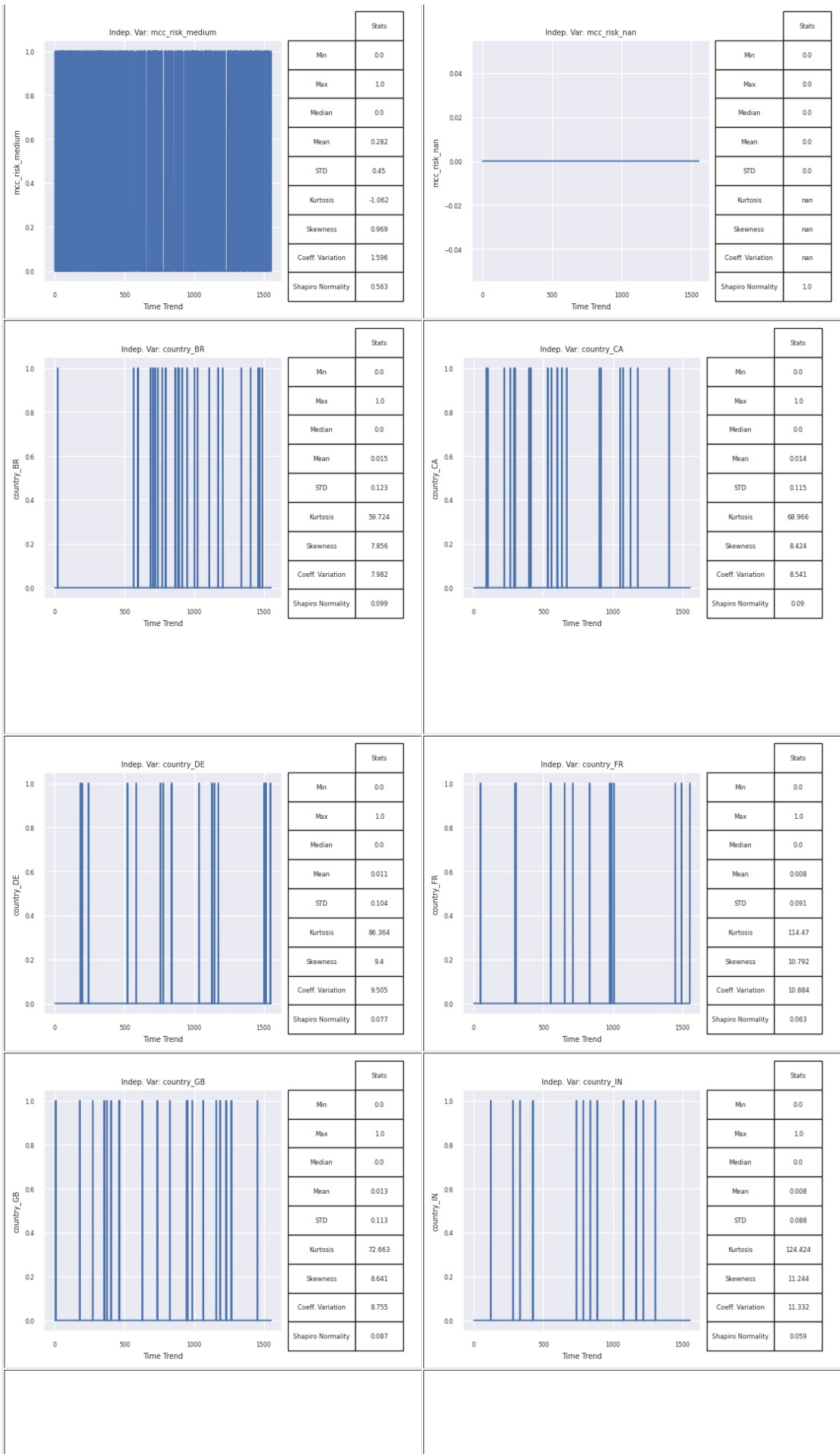
Detailed Graphs of Variables Against Time

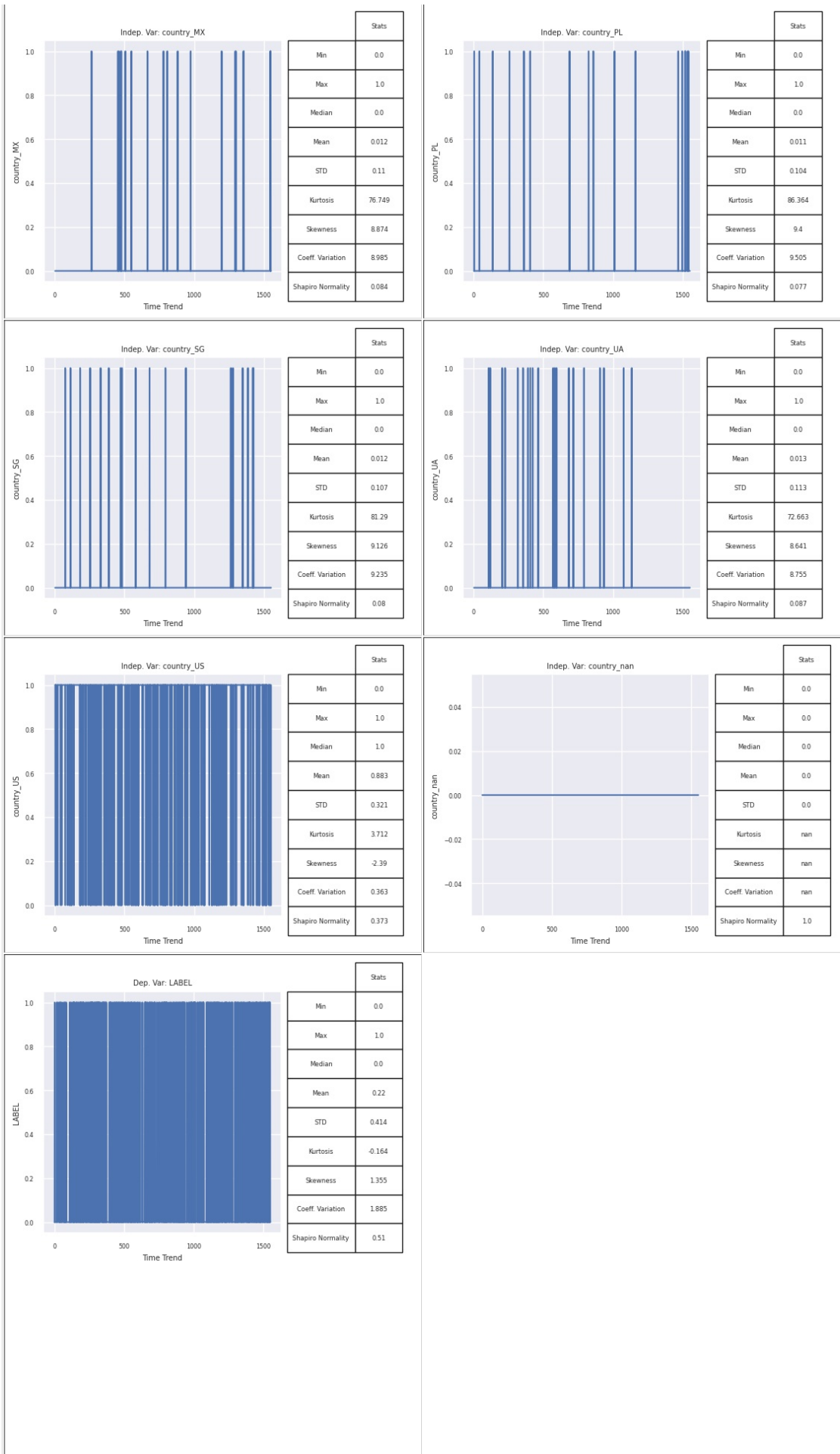




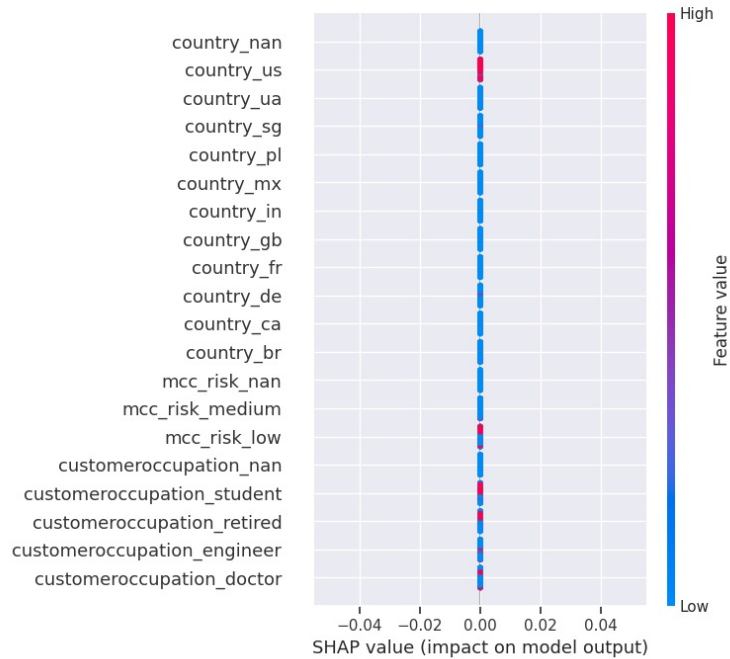








MODEL EXPLANATION



- The x-axis represents the model's output values of **LABEL**.
- The plot is centered on the x-axis at `explainer.expected_value`.
- All values are relative to the model's expected value like a linear model's effects are relative to the intercept.
- The y-axis lists the model's features. By default, the features are ordered by descending importance.
- The importance is calculated over the observations plotted. This is usually different than the importance ordering for the entire dataset.
- In addition to feature importance ordering, the decision plot also supports hierarchical cluster feature ordering and user-defined feature ordering.
- Each observation's prediction is represented by a colored line.
- At the top of the plot, each line strikes the x-axis at its corresponding observation's predicted value. This value determines the color of the line on a spectrum.
- Moving from the bottom of the plot to the top, SHAP values for each feature are added to the model's base value.
- This shows how each feature contributes to the overall prediction.
- At the bottom of the plot, the observations converge at `explainer.expected_value`.
- The points in the graph are the values of the feature in the training dataset.

FEATURE SELECTION	
RFE Variable (Most important to Least Important)	Value
unusual_mcc_for_customer	0.113
high_risk_mcc	0.076
proxy	0.031
mcc_risk_low	0.030
mcc	0.020
medium_risk_mcc	0.012
mcc_risk_medium	0.011
freq_2plus	0.009
customer_tenure_days	0.009
transactionduration	0.008
txn_velocity_5min_sim	0.008
customerage	0.008
high_frequency	0.006
txn_hour	0.005
country_US	0.005
channel_Online	0.004
channel_Branch	0.004
customeroccupation_Engineer	0.003
customeroccupation_Student	0.003
country_PL	0.003
customeroccupation_Doctor	0.003
transactiontype_Credit	0.002
transactiontype_Debit	0.002
customeroccupation_Retired	0.002
country_MX	0.002
loginattempts	0.002
country_CA	0.001
country_FR	0.001
short_tenure	0.001
country_IN	0.001
country_BR	0.001
country_SG	0.001
country_GB	0.001
country_DE	0.001
country_UA	0.001
new_ip	0.000
new_device	0.000
country_nan	0.000
customeroccupation_nan	0.000
mcc_risk_nan	0.000
channel_nan	0.000
transactiontype_nan	0.000
is_night	0.000
is_weekend	0.000
Best Variable(s) From Genetic Algorithm	
is_weekend	
Excluded Variable(s)	
unusual_mcc_for_customer	
transactionduration	
txn_velocity_5min_sim	
customerage	
high_frequency	
txn_hour	
country_US	
channel_Online	
channel_Branch	
customeroccupation_Engineer	
customeroccupation_Student	
high_risk_mcc	
country_PL	
customeroccupation_Doctor	
transactiontype_Credit	
transactiontype_Debit	
customeroccupation_Retired	
country_MX	
loginattempts	
country_CA	
country_FR	
short_tenure	
proxy	
country_IN	
country_BR	
country_SG	
country_GB	
country_DE	

country_UA	
new_ip	
new_device	
country_nan	
customeroccupation_nan	
mcc_risk_low	
mcc_risk_nan	
channel_nan	
transactiontype_nan	
is_night	
mcc	
medium_risk_mcc	
mcc_risk_medium	
freq_2plus	
customer_tenure_days	
PCA for Best Variable(s)	Value
is_weekend_pca_1	0.000
is_weekend_pca_2	1.000
PCA Explained Variance	Value
PCA1	1.000
PCA2	0.000
• Feature selection shows which variables were more influential than other variables • It uses two core algorithms: Recursive Feature Elimination (RFE) and Genetic Algorithm to determine influence • It also performs PCA (principal component analysis) analysis to determine the influence of the best variables in the model • These results should be used in conjunction with other information as well as theory to establish relevance and confidence in the chosen model formulation	

CLUSTER ANALYSIS: PCA and KMeans



Over 80% of the variance in your data is explained by **21** principal components.

Optimal clusters are **2** with a silhouette score of **0.372** using euclidean distance.

NOTE: Only two principal components are shown in the 2D graph because your variables have been reduced by dimensionality reduction using PCA.

However, the entire data set can be found here:

/maads/agentfilesdocker/dist/maadsweb/csvuploads/admin_maads_train_aug_binary_csv_clusters_pca_kmeans.csv for further analysis.

CORRELATION MATRIX



CORRELATED FEATURES

	Feature(s)	Feature(s)	Correlation >= 0.100
0	high_risk_mcc	channel_Online	0.104
1	freq_2plus	high_risk_mcc	0.106
2	channel_Online	high_frequency	0.113
3	high_risk_mcc	proxy	0.119
4	freq_2plus	country_BR	0.130
5	channel_Online	txn_velocity_5min_sim	0.137
6	country_DE	proxy	0.141
7	channel_Online	freq_2plus	0.153
8	channel_Online	transactiontype_Credit	0.162
9	proxy	channel_Online	0.226
10	country_UA	short_tenure	0.236
11	mcc	high_risk_mcc	0.564
12	customerage	customeroccupation_Retired	0.676
13	high_frequency	freq_2plus	0.686
14	txn_velocity_5min_sim	freq_2plus	0.859
15	high_frequency	txn_velocity_5min_sim	0.901
16	customerage	customerage	NaN

SUGGESTED CORRELATED FEATURES TO DELETE		
	4 Feature(s) to Delete	Correlation
	4 freq_2plus	0.130
	8 channel_Online	0.162
	10 country_UA	0.236
	12 customerage	0.676

END OF REPORT

MAADSBML MAIN DOCUMENTATION:<https://maadsbml.readthedocs.io/en/latest/>

MAADSBML Python Library:<https://pypi.org/project/maadsbml/>

MAADSBML Docker Container For Windows: <https://hub.docker.com/r/maadsdocker/maads-batch-automl-otics>

MAADSBML Docker Container For MAC: <https://hub.docker.com/r/maadsdocker/maads-batch-automl-otics-arm64>

MAADSBML Sample Code and Setup: <https://github.com/smaurice101/raspberrypi/tree/main/maadsbml>